SAT Msc Thesis Paper Template

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Nomenclature

Roman Letters

a, a_t	Action / Action vector at time step t
$\mathcal{A}$	Action space of the Goal-Conditioned Markov Decision Process
B	Number of bins in the spatial histogram grid
c_i	RECAT-EU wake turbulence category of aircraft i
C_{sep}	Overall inter-arrival separation compliance fraction
$d_{\text{go}}, \hat{d}, d_{\max}$	Actual distance-to-go / Estimated expected distance-to-go / Upper-bound of distance-to-go [m]
$d_{i,j}$	Distance between aircraft i and population grid cell j [m]
ΔD_{sep}	Minimum distance separation requirement [m]
D_{KL}	Kullback-Leibler divergence [bits]
E_i	Earliest possible landing time for aircraft i [s]
ETA_i	Estimated Time of Arrival for aircraft i [s]
$\hat{f}_{\text{ET}}$	Extra Trees regressor function
$g, \mathbf{g}$	Dynamic goal / Goal vector
$\mathcal{G}$	Total goal space (or Runway/Geographic group set)
$\mathcal{H}$	Spatial visitation entropy [bits]
$\mathcal{H}(\pi(\cdot))$	Policy Shannon entropy term
h_b	Histogram count in spatial grid bin b
J	Expected return objective function for reinforcement learning
k	Maximum position shifting bound in Constrained Position Shifting
L	Total trajectory path length [m]
ℓ_i	Cumulative path length at step i [m]
L_i	Latest possible landing time for aircraft i [s]
$\mathcal{L}$	Loss function for critic or actor networks
$\mathcal{M}$	Goal-Conditioned Markov Decision Process tuple
M	Total number of evaluation episodes
M_g	Total number of evaluation episodes for group g

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M_{update}	Total number of evaluation episodes experiencing a dynamic target change
N_a	Total number of aircraft in an arrival sequence
$N_{a,r}$	Total number of aircraft assigned to runway r
$N_{\text{success},r}$	Count of successfully coordinated aircraft on runway r
$\mathcal{N}$	Cumulative community noise reward
P	Set of all approaching aircraft
p_b	Discrete probability of spatial grid bin b
P_j	Population count of grid cell j
$\mathbf{p}_0$	Initial aircraft spawn position vector
$\mathbf{p}^*$	Target runway threshold position vector
$\mathcal{P}$	Transition probability function of the GC-MDP
Q_ϕ	Goal-conditioned action-value function parametrised by ϕ
R	Reward function component (or total number of runways)
$\mathcal{R}$	Reward function space of the GC-MDP (or candidate runway set)
R_{rec}	Empirical recovery success rate fraction
s, s_t	Environment state vector / State vector at time step t
S_{ij}	Same-runway minimum separation requirement [s]
s_{ij}	Different-runway minimum separation requirement [s]
$\mathcal{S}$	State space of the GC-MDP
t	Elapsed simulation time / current planning epoch [s]
T	Total episode horizon or terminal step [s]
$\hat{T}_i$	Predicted remaining flight time until IAF crossing [s or simulation steps]
$T_{\max}$	Maximum simulation duration / time normalization factor [s]
ΔT_{plan}	Strategic trajectory re-planning interval [s or simulation steps]
ΔT_{sep}	Minimum time separation requirement [s]
$T_{\text{seq},r}$	Total elapsed duration of the arrival sequence on runway r [s]
TTA_i	Target Time of Arrival for aircraft i [s]
$\mathcal{T}$	Path tortuosity index
V	Set of aircraft pairs requiring enforced separation
v_{app}	Fixed standard approach speed [m/s]
w_{pop}	Population noise weighting factor
x_i	Assigned landing time for aircraft i [s]
X	Uniformly distributed random variable for slack generation
y_{ir}	Binary runway allocation indicator variable
y_t	Bellman optimisation target at time step t
z_{ij}	Binary shared runway assignment indicator variable

Greek Letters

α	Entropy regularisation temperature coefficient
γ	Discount factor for future rewards
Γ	Arrival throughput [aircraft/hr]
δ_{ij}	Binary landing sequence indicator variable
δ_{slack}	Stochastic delay slack term [m]
δ_t	Operational temporal tolerance threshold [s]
δ_{update}	Update tolerance threshold for TTA modifications [s]
$\Delta\epsilon^{(m)}$	Tracking error degradation for episode m [s]
ϵ_{RTA}	Required Time of Arrival tracking error [s or dimensionless]
$\bar{\epsilon}_{\text{RTA}}$	Mean Required Time of Arrival tracking error across a sequence [s]
$\tilde{g}$	Hindsight experience replay pseudo-goal vector

ρ_{ripple}	Delay ripple index
σ_i	Sequential position index of aircraft i
τ	Flight time duration [s]
τ^*	Normalised Required Time of Arrival goal component
$\hat{\tau}$	Actual realised arrival time [s]
ξ	Normalised episode progress at termination
π	Policy function parametrised by actor network weights
ϕ	Fraction threshold of peak success rate / Critic network parameter index
ψ	Spatial aircraft spawn bearing [deg]

Subscripts and Superscripts

aug	Augmented state/reward formulation
b	Spatial grid bin index
dynamic	Mid-trajectory target updated configuration
g	Geographic group index
go	Remaining distance identifier
i, j	Aircraft index identifiers
(m)	Evaluation episode index identifier
max	Operational or simulation ceiling limit
ϕ	Critic network parameter index
plan	Strategic re-planning identifier
r	Runway designator index
rec	Empirical recovery identification
ripple	Delay ripple propagation component
sep	Wake turbulence separation constraint identifier
sparse	Sparse reward signal component
static	Baseline unrevised configuration
t	Time step index

miscellaneous

$\mathbb{1}$	Indicator function

I. Introduction (Deliverable for the Mid-Term meeting at 15 weeks)

The modernisation of Air Traffic Management (ATM) is reaching a critical inflection point. As global air traffic volumes recover from the COVID-19 pandemic and expand, with passenger volumes projected by the International Civil Aviation Organisation (ICAO) to nearly triple to 12.4 billion by 2050 [1], Terminal Control Areas (TMAs) have become the primary bottlenecks of the global airspace network. Traditionally, the tasks of arrival sequencing, runway assignment and 4D-trajectory adherence have relied on a combination of manual, experience-based vectoring and rigid, rule-based scheduling. However, the increasing complexity of modern airspace, driven by rising demand, the integration of new entrants such as zero-emission aircraft and stringent environmental mandates, requires a transition toward Trajectory Based Operations (TBO) [2, 3].

Recent breakthroughs in Reinforcement Learning (RL) have demonstrated that autonomous agents can successfully learn noise-optimal and fuel-efficient trajectories in the TMA. Specifically, the hierarchical model developed by Groot et al. [4] proved that agents can navigate complex airspace with high precision, achieving reductions in separation violations of over 99%. However, a core limitation of this framework is that it treats the airport as a single "sink" with infinite capacity, optimising only for path-based metrics. In multi-runway environments, this produces the "Runway Overload" phenomenon: multi-agent systems converge on a single efficient route, leaving other runways underutilised and violating inter-arrival time constraints.

Traditional Aircraft Landing Problem (ALP) formulations, often based on Mixed-Integer Linear Programming (MILP) or heuristic and metaheuristic methods, address the complementary challenge by providing rigorous temporal scheduling and wake-turbulence separation guarantees. However, these methods treat spatial trajectory generation as exogenous, relying on fixed speed profiles and predefined routes. While exact methods provide high fidelity for benchmarking, they are fundamentally NP-hard and thus struggle with the computational scalability required for high-density, real-time scenarios [5, 6]. Hybrid heuristic approaches, such as Constrained Position Shifting (CPS) [7], demonstrate that near-optimal throughput can be achieved with linear scaling, yet they remain largely decoupled from continuous tactical aircraft control. Similarly, although RL offers powerful tools for high-dimensional state spaces, Razzaghi et al. [8] highlight that most implementations remain task-specific and lack integration into operationally coherent TMA-wide frameworks.

Recognising this disconnect, recent research has proposed frameworks to jointly consider the spatial and temporal dimensions of arrival management. Louie et al. [9] introduced the Integrated Arrival Solution (IAS), which utilises a "trajectory grafting" methodology to make landing order a by-product of trajectory feasibility rather than a pre-imposed constraint. While IAS reduces fuel consumption and demonstrates that decoupled processes are fundamentally unpredictable in continuous dynamic operations, its sequential search process introduces significant computational overhead and its greedy, fuel-centric selection can lead to sub-optimal global runway utilisation, a limitation shared by Groot et al. [4]. Wang et al. [10] addressed the "System Integration Gap" through the MSTAGNN-MARL framework, which represents air traffic as a spatial-temporal graph for time-slot allocation using Graph Neural Networks (GNNs). Despite these advances, the framework assumes simplified 2D flight dynamics and exhibits performance degradation under extreme traffic density. A persistent "Spatio-Temporal Gap" therefore remains: the structural disconnect between strategic Arrival Managers (AMAN), which calculate abstract timestamps, and tactical flight execution, which must handle full aircraft dynamics in continuous time.

The main contribution of this paper is the formalisation of the Trajectory Based Aircraft Landing Problem (TBALP) and the development of a Goal-Conditioned Hierarchical Reinforcement Learning (GCHRL) framework to address it, directly extending the path-planning model of Groot et al. [11] into a full arrival management system. A high-level Manager, implemented as a CPS scheduler that restricts each aircraft to within k positions of its First-Come-First-Served order [7], handles runway allocation and assigns Required Times of Arrival (RTAs) drawn from an introduced curriculum of varying temporal pressure. Concurrently, a low-level "4D-Worker" is trained using Soft Actor-Critic (SAC) [12] to treat time as a primary dimension of its state-action space. This enables emergent tactical manoeuvres such as path-stretching to satisfy RTA constraints without pre-defined geometric templates. To ensure spatio-temporal consistency, an Extra Trees regressor [13], following the lightweight surrogate modelling approach proposed by Dhief et al. [14], serves as a functional bridge, providing the Manager with rapid Estimated Time of Arrival (ETA) approximations without requiring a full trajectory simulation. The framework is evaluated within BlueSky-gym [15], an open-source air traffic simulator that unifies BlueSky [16] with a standardised RL interface, providing a high-fidelity environment for testing generalisation to unforeseen scheduling constraints.

This paper is outlined as follows: section II introduces the baseline hierarchical RL model, the Aircraft Landing Problem and its classical solution methods, goal-conditioned hierarchical reinforcement learning, and lightweight surrogate modelling for spatio-temporal bridging; section III presents the GCHRL framework, comprising the CPS manager and ETA surrogate, the two-stage spatial-to-temporal training procedure, and the single-agent evaluation framework; section IV details the BlueSky-gym experimental setup; section V presents an analysis of emergent tactical behaviours and framework performance under varying traffic densities; section VII concludes with a summary of findings regarding RTA precision and directions for future extensions to the strategic manager.

II. Background (Deliverable for the Mid-Term meeting at 15 weeks)

This section situates the proposed GCHRL framework within the broader Air Traffic Management (ATM) context relative to the multi-agent baseline of Groot et al. [4] and three foundational bodies of literature. Specifically, a review is presented of the baseline model architecture, the Aircraft Landing Problem (ALP) and its classical solution methods, goal-conditioned and hierarchical reinforcement learning, and lightweight surrogate modelling for spatio-temporal bridging.

A. Baseline Hierarchical RL Model

The framework proposed in this paper directly extends the multi-agent hierarchical RL model of Groot et al. [4], which decomposes terminal area control into four interdependent modules. A strategic Path Planner generates lateral headings by solving a Markov Decision Process (MDP) with a bi-criterion reward, balancing noise exposure and fuel consumption over the approach. A rule-based Vertical Control module maintains a 3° glideslope analytically, while two tactical RL modules handle multi-agent separation (using an additive attention mechanism over neighbouring aircraft states) and traffic stream merging respectively. Together these achieve a reduction in separation intrusions of over 99% in the Dutch TMA. **While the full architecture is acknowledged, the extension developed in this work focuses exclusively on the strategic Path Planner module.** Currently, this module treats the airport as a single fixed sink: the path planner's reward is purely path-based, there is no runway assignment logic and inter-arrival timing is managed only reactively through local speed adjustments and path stretching strategies imposed by the separation modules. The three subsections below introduce the bodies of literature required to extend this specific baseline component into a full arrival management system: the ALP and its sequencing constraints, the goal-conditioned and hierarchical RL machinery that makes the Worker temporally aware, and the surrogate modelling that bridges the two layers.

B. The Aircraft Landing Problem and Wake-Turbulence Separation

The ALP is a combinatorial optimisation problem focused on sequencing and scheduling arrivals at a runway system to maximise throughput while maintaining inter-aircraft safety margins. In its canonical MILP formulation [17], each aircraft $i \in P$ is assigned a landing time $x_i \in [E_i, L_i]$, a binary landing sequence variable $\delta_{ij} \in \{0, 1\}$ and a binary runway assignment $y_{ir} \in \{0, 1\}$, with the objective of minimising weighted deviation from target landing times subject to sequence-dependent separation constraints applied to the set V of aircraft pairs requiring enforced separation:

$$x_j \geq x_i + S_{ij} z_{ij} + s_{ij}(1 - z_{ij}) \quad \forall\, (i, j) \in V \tag{1}$$

where S_{ij} and s_{ij} denote same- and different-runway separation minima respectively, and z_{ij} indicates shared runway assignment. Although MILP provides an optimality benchmark, the ALP is NP-hard; computational complexity grows exponentially with aircraft count, rendering exact methods intractable for real-time, high-density multi-runway operations [17].

To address scalability, Balakrishnan et al. [7] demonstrated that CPS, which restricts each aircraft to within k positions of its First-Come-First-Served (FCFS) order, achieves near-optimal throughput under Dynamic Programming with linear scaling. CPS forms the basis of the strategic Manager layer in this work. The separation matrix it enforces adopts the **RECAT-EU** standard [18], which partitions aircraft into six categories (A–F) by wingspan and maximum take-off mass, providing a more granular distance-based separation scheme than legacy ICAO categories and reducing unnecessary over-separation between medium and heavy aircraft.

A persistent limitation of classical ALP methods, including both exact and heuristic approaches [6], is that spatial trajectory generation is treated as exogenous: landing time windows $[E_i, L_i]$ are fixed parameters derived from simplified performance assumptions rather than from a live trajectory model. In continuous dynamic operations, these windows are intrinsically coupled to the aircraft's current speed profile and path geometry, motivating the surrogate modelling approach described in subsection II.D.

C. Goal-Conditioned Hierarchical Reinforcement Learning

Standard RL trains a policy $\pi(a|s)$ to maximise the expected discounted return $\mathbb{E}[\sum_t \gamma^t R_t]$ within an MDP [19], which is insufficient for multi-runway arrival management, where the destination and timing constraint are dynamically assigned by a higher-level scheduler. This work formulates the tactical execution problem as a **Goal-Conditioned Markov Decision Process (GC-MDP)**, defined by the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma, \mathcal{G})$, in which the policy $\pi(a|s, g)$ is conditioned on a dynamic goal $g \in \mathcal{G}$ representing an assigned runway threshold combined with an RTA. Universal Value Function Approximators (UVFA) [20] enable the agent to generalise across the full goal space without per-goal retraining.

Embedding GCRL within a hierarchical structure following the manager-worker paradigm of Vezhnevets et al. [21] and the options framework of Sutton et al. [22] separates temporal and functional scales: the Manager operates at a strategic horizon to issue goal-conditioned TTAs, while the 4D-Worker executes continuous tactical control at each time step. This decomposition directly addresses the "Runway Overload" failure mode identified by Groot et al. [4], where a flat path-planning policy converges on a single noise-optimal route regardless of runway capacity.

The 4D-Worker is trained using Soft Actor-Critic (SAC) [12], an off-policy actor-critic algorithm that augments the standard expected-return objective with a Shannon entropy term:

$$J(\pi) = \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{T} \gamma^t \left(R(s_t, a_t, g) + \alpha\,\mathcal{H}(\pi(\cdot|s_t, g))\right)\right] \tag{2}$$

where $\alpha > 0$ is a temperature coefficient that controls the exploration–exploitation trade-off and $\mathcal{H}(\pi(\cdot|s, g)) = -\mathbb{E}_{a\sim\pi}[\log \pi(a|s, g)]$ is the policy entropy. The temperature coefficient dynamically scales the policy's entropy relative to the reward. Operationally, this dictates the structural variety of the generated trajectories: a higher entropy weight forces the spatial Worker to actively explore alternate heading variations and path-stretching manoeuvres within the TMA boundaries, mapping out the full physical boundaries of the arrival-time envelope rather than converging prematurely on a single rigid track. This is critical in the TBALP where path-stretching and speed adjustment must remain available across the full RTA distribution.

The composite reward decomposes into a dense augmented component R_{aug} that accumulates path-length and noise penalties at each step, and a sparse terminal component R_{sparse} that delivers a goal-attainment signal on episode completion. SAC maintains two independent critic networks Q_{ϕ_1}, Q_{ϕ_2} to estimate the goal-conditioned action-value function, and uses the minimum of their outputs to form conservative Bellman targets, reducing overestimation bias [12]:

$$y_t = R(s_t, a_t, g) + \gamma \min_{i=1,2} Q_{\phi_i}(s_{t+1}, \tilde{a}_{t+1}, g) - \alpha \log \pi(\tilde{a}_{t+1}|s_{t+1}, g), \quad \tilde{a}_{t+1} \sim \pi \tag{3}$$

Each critic is updated by minimising the mean squared Bellman error $\mathcal{L}(\phi_i) = \mathbb{E}[(Q_{\phi_i}(s_t, a_t, g) - y_t)^2]$, while the Actor is updated by minimising $\mathcal{L}(\pi) = \mathbb{E}_{s,g}[\alpha \log \pi(a|s, g) - Q_\phi(s, a, g)]$.

The central learning challenge is that temporal goal-conditioning introduces sparse rewards in a high-dimensional state space: the agent rarely lands within a tight RTA window during early training. **Hindsight Experience Replay (HER)** [23] resolves this by re-labelling failed trajectories post hoc. Given a trajectory $\{(s_t, a_t, g)\}_{t=0}^{T}$ that failed to satisfy $g = (p^*, \tau^*)$, HER substitutes a pseudo-goal $\tilde{g} = (p_T, \tau_T)$ derived from the actually reached terminal state, transforming every transition into a valid learning signal:

$$(s_t,\ a_t,\ R(s_t, a_t, g),\ s_{t+1},\ g) \longrightarrow (s_t,\ a_t,\ R(s_t, a_t, \tilde{g}),\ s_{t+1},\ \tilde{g}) \tag{4}$$

This re-labelling teaches the agent the precise path geometry required to realise arrival time $\hat{\tau}$, even when τ^* was unachievable. Crucially, re-labelling is only effective when $\tilde{g}$ falls within the agent's reachable set. While HER addresses goal re-labelling, it does not account for temporal variations. Consequently, a stochastic RTA sampler grounded in the spatial policy's empirical path-length distribution must be introduced to ensure the replay buffer spans the full achievable arrival-time envelope. This empirical path-length distribution is, in essence, a precursor to the surrogate modelling approach described in subsection II.D: both regress aircraft state onto trajectory-derived quantities.

D. Surrogate Temporal Modelling for Spatio-Temporal Bridging

A structural "chicken-and-egg" dependency exists between the Manager and the Worker: the Manager cannot assign a feasible RTA without knowing when the aircraft can realistically land, yet executing a full trajectory rollout to obtain that estimate is computationally prohibitive at scheduling frequency. This work resolves the dependency through a lightweight ETA surrogate model, following the approach of Dhief et al. [14], who demonstrated that tree-based ensembles significantly outperform physics-based baselines in both prediction speed and accuracy for terminal-area arrival estimation.

Concretely, an **Extra Trees (ET) regressor** [13] is trained on rollouts of the spatial policy to map the current aircraft state to a predicted ETA at the runway threshold. Unlike standard Random Forests that optimise splits deterministically across all features, the Extra Trees regressor draws split thresholds uniformly at random. In the context of terminal airspace modelling, this structural randomisation prevents the surrogate from overfitting to localised, patterns along specific arrival corridors, thereby stabilising the continuous ETA estimation without inflating downstream scheduling variance. The ET surrogate provides the CPS Manager with a feasible temporal window for RTA assignment in near real time, removing the need for a full trajectory simulation at each re-planning step. The resulting Manager-surrogate-Worker loop constitutes the functional TBO–ALP bridge at the core of the GCHRL framework.

III. Methodology

The methodology is structured top-down, from strategic coordination to tactical execution. The CPS Manager and its ETA surrogate are defined first, establishing the temporal constraints and goal-conditioning interface for the worker agent (subsection III.B). The two-stage spatial-to-temporal RL framework follows, covering HER-based re-labelling and stochastic RTA sampling from the empirical path-length distribution (subsection III.C). Single-agent evaluation metrics are then formalised, applied uniformly across all ablations (subsection III.D). Training data dependencies are inverted: the spatial policy core must be trained and frozen before the strategic surrogates can be fitted.

A. Scope and Core Assumptions

Three operational assumptions bound the system:

1) **Kinematic Uniformity:** A homogeneous A320 fleet operates at constant calibrated airspeed (v_{app}); the 4D-Worker absorbs scheduling delays exclusively via lateral path-stretching.
2) **Static Runway Assignment:** Except where the dynamic extension is evaluated, runway and IAF are committed on TMA entry based on arrival sector, with no subsequent reassignment.
3) **Environmental Determinism:** Wind, weather corridors and turbulence are omitted to isolate the coupling between the RL policy and the scheduling manager.

The dependent measures arising from these assumptions are summarised in Table 1.

Table 1 Summary of dependent measures within the TBALP framework, categorised by operational evaluation domain.

Evaluation Domain & Metric	Operationalism	Operational Function & Purpose
System Coordination & Throughput (CPS Manager)		
Throughput (Γ)	Equation III.B.4	Evaluates absolute and per-runway capacity (IAF crossings per unit time).
Separation Compliance (C_{sep})	Equation 11	Quantifies operational safety against RECAT-EU wake turbulence separation minima.
Tracking Degradation ($\Delta\epsilon$)	Equation 12	Measures tactical precision loss induced by mid-trajectory dynamic TTA target updates.
Recovery Success Rate (R_{rec})	Equation 13	Evaluates macro-level system resilience as the fraction of updated flights meeting tolerance δ_t.
Delay Ripple Index (ρ_{ripple})	Equation 14	Monitors downstream perturbation cascading via lag-1 autocorrelation of RTA errors.
Tactical Performance & Trajectory Quality (Worker Agent)		
Success Rate (S, S_g)	Equation 22	Primary metric tracking simultaneous spatial (SINK) and temporal (δ_t) goal attainment.
RTA Error (ϵ_{RTA})	Equation 23	Quantifies arrival precision and identifies directional bias (early vs. late arrivals).
Efficiency (L, τ)	subsubsection III.D.2	Logs total flight path length and elapsed time under constant-velocity assumptions.
Path Tortuosity ($\mathcal{T}$)	Equation 24	Quantifies trajectory detours caused by lateral path-stretching or noise avoidance behaviour.
Environmental Impact & Spatial Distribution		
Noise Exposure ($\mathcal{N}$)	subsubsection III.D.2	Computes cumulative community noise exposure over populated geographic grids.
Spatial Visitation Entropy ($\mathcal{H}$)	Equation 25	Measures geographic routing diversity and airspace coverage density.
KL Divergence (D_{KL})	Equation 26	Quantifies macro-level routing strategy shifts between distinct policy configurations.
Training Efficiency & Learning Dynamics		
Reward Area Under the Curve (AUC)	Equation 19	Summarises overall learning speed and sample efficiency across training updates.
Steps to Threshold (t_ϕ)	Equation 20	Tracks environment steps needed to reach stable fractions (ϕ) of peak performance.

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Table 1 – continued from previous page

Evaluation Domain & Metric	Operationalism	Operational Function & Purpose
Tail Coeff. of Variation (CV_{tail})	Equation 21	Assesses asymptotic policy stability and flags post-peak performance regression.

B. Constrained Position Shifting Manager

The CPS manager coordinates a sequence of N_a aircraft approaching the TMA, assigning a Target Time of Arrival (TTA) at the IAF and reissuing updates at fixed re-planning intervals ΔT_{plan}, using only the ETA estimate from the surrogate defined in subsubsection III.B.1. Three time references are distinguished: the ETA is the surrogate's predicted natural IAF arrival; the TTA is the manager-assigned schedule, which may deviate from the ETA to enforce separation; and the RTA is the normalised TTA presented to the agent as its temporal goal.

1. ETA Surrogate Construction

An Extra Trees regressor [14] trained on empirical rollouts of the frozen spatial policy serves as the ETA surrogate. Unlike the training-stage DTG surrogate, which predicts remaining path distance, the ETA surrogate predicts $\hat{T}_i$, the simulation steps remaining until IAF crossing:

$$\hat{T}_i = \hat{f}_{\text{ET}}(\cdot) \tag{5}$$

where the feature vector follows the coordinate system and temporal conditioning selected in subsubsection III.C.2. Predictive accuracy (R^2, MAE, RMSE) is reported on a held-out validation set prior to deployment.

2. Sequence Management and TTA Assignment

At each planning epoch at elapsed time t, the manager maintains an ordered arrival sequence $O = \langle a_1, a_2, \ldots, a_{N_a} \rangle$ ranked by absolute estimated arrival times:

$$\text{ETA}_i = t + \hat{T}_i \tag{6}$$

This ranking defines the First-Come-First-Served (FCFS) reference order, from which each aircraft may deviate by at most k positions under the Constrained Position Shifting constraint [7]. Runway assignment is determined on TMA entry based on arrival direction, fixing the IAF and applicable RECAT-EU separation standard.

A dynamic assignment extension treats runway allocation as a decision variable, reassigning aircraft to the runway yielding the earliest feasible TTA subject to separation constraints:

$$r_i^* = \arg\min_{r \in \mathcal{R}} \ \text{TTA}_i^{(r)} \quad \text{subject to} \quad \left| \sigma_i^{(r)} - \sigma_i^{\text{FCFS}} \right| \leq k \tag{7}$$

where $\sigma_i^{(r)}$ is the projected sequential position index of aircraft a_i on runway r and σ_i^{FCFS} its baseline FCFS rank. This balances runway load without relaxing fairness constraints and requires no architectural changes to the Worker or ETA surrogate.

Separation is enforced per RECAT-EU wake turbulence categories [18], converting minimum distance separation $\Delta D_{\text{sep}}(c_i, c_{i+1})$ to a minimum time gap:

$$\Delta T_{\text{sep}}(c_i, c_{i+1}) = \frac{\Delta D_{\text{sep}}(c_i, c_{i+1})}{v_{\text{app}}} \tag{8}$$

TTAs are assigned greedily forward subject to the k-CPS constraint:

$$\text{TTA}_1 = \text{ETA}_1, \qquad \text{TTA}_i = \max\big(\text{ETA}_i,\ \text{TTA}_{i-1} + \Delta T_{\text{sep}}(c_{i-1}, c_i)\big)\,, \quad i = 2, \ldots, N_a \tag{9}$$

The assigned TTA is normalised to the agent's temporal goal as $\tau^* = \frac{\text{TTA}_i}{T_{\max}}$.

3. Dynamic TTA Updates

At each re-planning epoch, ETAs are recomputed via Equation 5, the sequence is re-ranked and TTAs are reassigned via Equation 9. An agent's TTA is updated only if the new assignment differs from the previous by more than δ_{update}, preventing spurious re-targeting. The updated TTA is injected mid-trajectory as $\tau^* = \frac{\text{TTA}_i}{T_{\max}}$ without resetting the episode; the agent receives no explicit revision signal, so adaptation must emerge from the goal-conditioned policy via the Markov property.

4. Throughput and Arrival Precision Metrics

Performance is evaluated over a full arrival sequence of N_a aircraft across R runways. Per-runway throughput is:

$$\Gamma_r = \frac{N_{\text{success},r}}{T_{\text{seq},r}} \tag{10}$$

where $N_{\text{success},r}$ counts aircraft on runway r with $\left|\epsilon_{\text{RTA}}^{(m)}\right| \leq \delta_t$ and $T_{\text{seq},r}$ is the elapsed time from first to last IAF crossing on that runway. Total throughput is $\Gamma = \sum_{r=1}^{R} \Gamma_r$ (). Per-runway separation compliance is:

$$C_{\text{sep},r} = \frac{1}{N_{a,r} - 1} \sum_{i=1}^{N_{a,r}-1} \mathbb{1}\left[\hat{T}_{i+1} - \hat{T}_i \geq \Delta T_{\text{sep}}(c_i, c_{i+1})\right] \tag{11}$$

with overall compliance $C_{\text{sep}} = \frac{\sum_{r=1}^{R}(N_{a,r}-1)\cdot C_{\text{sep},r}}{\sum_{r=1}^{R}(N_{a,r}-1)}$. Both metrics are compared against an uncoordinated baseline and against Groot et al. [4] as detailed in the experimental setup.

Tracking degradation under dynamic re-planning is:

$$\Delta\epsilon^{(m)} = \left|\epsilon^{(m)}_{\text{RTA, dynamic}}\right| - \left|\epsilon^{(m)}_{\text{RTA, static}}\right| \tag{12}$$

Macro-level resilience is captured by the recovery success rate:

$$R_{\text{rec}} = \frac{1}{M_{\text{update}}} \sum_{m=1}^{M_{\text{update}}} \mathbb{1}\left[\left|\epsilon^{(m)}_{\text{RTA, dynamic}}\right| \le \delta_t\right] \tag{13}$$

where M_{update} is the number of episodes with mid-trajectory goal changes. The delay ripple index monitors downstream disruption via lag-1 autocorrelation of signed arrival errors [24]:

$$\rho_{\text{ripple}} = \frac{\sum_{i=1}^{N_a-1}(\epsilon_{\text{RTA},i} - \bar{\epsilon}_{\text{RTA}})(\epsilon_{\text{RTA},i+1} - \bar{\epsilon}_{\text{RTA}})}{\sum_{i=1}^{N_a}(\epsilon_{\text{RTA},i} - \bar{\epsilon}_{\text{RTA}})^2} \tag{14}$$

where $\bar{\epsilon}_{\text{RTA}}$ is the sequence mean. Lag-1 is appropriate because wake turbulence constraints are pairwise; a high positive ρ_{ripple} indicates cascading errors, while a value near or below zero indicates effective perturbation isolation.

C. Spatial-to-Spatial-Temporal Extension

The 4D agent is developed in two stages. In Stage I, the agent trains in a purely spatial setting with state $\langle x, y, t\rangle$ and goal $\mathbf{g} = \langle g_x, g_y, \tau^*\rangle$, with temporal information suppressed by fixing $\tau^* = 0$. The agent is rewarded only for spatial goal satisfaction, path length and noise exposure, yielding a spatially competent policy whose rollouts provide the empirical path-length distribution required for Stage II. In Stage II, a DTG sampler is fitted to those rollouts and temporal goals are injected to activate the full 4D problem. The two-stage pipeline is illustrated in Figure 1.

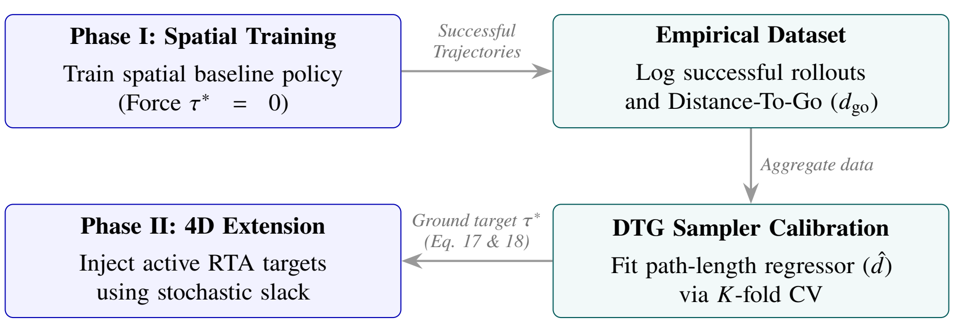


Fig. 1 Two-stage spatial-to-spatial-temporal training pipeline architecture.

1. RTA Data Collection

The spatial policy is rolled out for N successful episodes, recording $(x_i, y_i, t_i, \text{runway}, \ell_i)$ at each step, where ℓ_i is cumulative path length in km. Distance-to-go at each step is:

$$d_{\text{go},i} = L - \ell_i \quad [\text{km}] \tag{15}$$

2. DTG Sampler Construction

The DTG sampler ensures training RTA targets are reachable under the spatial policy; without it, goals would fall outside the agent's achievable set and deprive it of positive reward signal. At deployment, the RTA is assigned by the CPS manager, so sampler-based goal construction is a training-only mechanism.

Two formulations are considered, estimating expected remaining path distance from a given state:

$$\begin{aligned} \hat{d}_{\text{s}} &= \mathbb{E}\left[d_{\text{go}} \mid x, y, \text{runway}\right] \\ \hat{d}_{\text{st}} &= \mathbb{E}\left[d_{\text{go}} \mid x, y, t, \text{runway}\right] \end{aligned} \tag{16}$$

Candidate estimators are evaluated across Cartesian (x, y) and polar (r, θ) representations via K-fold cross-validation using R^2, MAE, RMSE and sMAPE [25]. Significance of performance differences is assessed via paired Diebold-Mariano (DM) [26] and Wilcoxon signed-rank [27] tests on cross-validated per-sample errors; a difference is accepted only when both agree at $p < 0.05$. The statistical foundations of these tests and the rationale for their combined use are given in section VII. Final selection is based on cross-validated R^2 and RMSE, with the tests confirming the margin is not due to chance.

3. Temporal Goal Injection

In Stage II, the DTG sampler constructs τ^* at each episode reset from the sampled runway, spawn position (x_0, y_0) and, if using the temporally-augmented formulation, $t_0 = 0$. A stochastic slack term encourages exploration across the temporal envelope:

$$\delta_{slack} = (d_{\max} - \hat{d}) \cdot X, \qquad X \sim \mathcal{U}(0, 1) \tag{17}$$

where $d_{\max}$ is a configurable upper bound on the DTG, bounding slack to $[0,\ d_{\max} - \hat{d}]$. The normalised RTA goal is:

$$\tau^* = \frac{\hat{d} + \delta_{slack}}{v \cdot T_{\max}} \tag{18}$$

where v is aircraft speed and $T_{\max}$ normalises to the observation space.

D. Single-Agent Model Evaluation

Each configuration is evaluated along three axes: training efficiency, evaluation performance and behavioural quality, applied uniformly across all ablations including HER versus No-HER, state-space variations and the spatial-to-temporal extension.

1. Training Efficiency

Three scalars are derived from reward and success rate curves recorded at evaluation intervals $\{t_1, \ldots, t_N\}$ over T total environment steps.

Area Under the Mean Reward Curve.

$$\mathrm{AUC} = \sum_{n=1}^{N-1} \frac{\bar{R}(t_n) + \bar{R}(t_{n+1})}{2} \cdot (t_{n+1} - t_n) \tag{19}$$

Steps to Threshold Success Rate.

$$t_\phi = \min\{t_n : S(t_n) \geq \phi \cdot S^*\}, \quad \phi \in \{0.80,\ 0.90\} \tag{20}$$

where $S(t_n) \in [0, 1]$ is the empirical success rate at step t_n and S^* the peak success rate.

Tail Coefficient of Variation. Late-training stability is measured over the tail window $\mathcal{T} = \{t_n : t_n > \tau T\}$, where $\tau = 0.8$:

$$\mathrm{CV}_{\mathrm{tail}} = \frac{\sigma_{\mathcal{T}}}{\mu_{\mathcal{T}}}, \quad \mu_{\mathcal{T}} = \frac{1}{|\mathcal{T}|} \sum_{t_n \in \mathcal{T}} \bar{R}(t_n), \quad \sigma_{\mathcal{T}} = \sqrt{\frac{1}{|\mathcal{T}|} \sum_{t_n \in \mathcal{T}} \left(\bar{R}(t_n) - \mu_{\mathcal{T}}\right)^2} \tag{21}$$

The peak-to-final drop $\Delta_{pf} = \bar{R}^* - \bar{R}(T)$, where $\bar{R}^* = \max_n \bar{R}(t_n)$, additionally flags reward regression after the best checkpoint.

2. Evaluation Performance

Each agent is evaluated over M episodes spanning all runway groups $\mathcal{G}$, with spawn positions and (where applicable) RTA sampled uniformly from the operational envelope.

Per-Group Success Rate. For spatial-temporal configurations, success requires simultaneous spatial goal satisfaction and arrival within tolerance δ_t:

$$S_g = \frac{1}{M_g} \sum_{m=1}^{M_g} \mathbb{1}\left[s_k^{(m)} \in \mathrm{SINK} \wedge \left|\epsilon_{\mathrm{RTA}}^{(m)}\right| \leq \delta_t\right], \qquad S = \frac{\sum_{g \in \mathcal{G}} M_g \cdot S_g}{\sum_{g \in \mathcal{G}} M_g} \tag{22}$$

where M_g is the episode count for group g and $s_k^{(m)}$ the terminal state; the temporal constraint is omitted for spatial-only configurations.

Operational Metrics. Per successful episode, path length $L^{(m)}$ (2D arc length in km), flight time $\tau^{(m)}$ and cumulative noise exposure $\mathcal{N}^{(m)} = \sum_{i=0}^{k} w_{\text{pop}} \cdot \sum_{j \in \text{pop}} \frac{P_j}{d_{i,j}^2}$ are recorded. Per-group means and standard deviations are reported.

Temporal Precision. Arrival precision is quantified by the RTA error at the terminal step:

$$\epsilon_{\text{RTA}}^{(m)} = \hat{\tau}^{(m)} - \tau^* \tag{23}$$

where $\hat{\tau}^{(m)}$ is actual arrival time and τ^* the RTA, both normalised. Error distributions and per-RTA-bin success rates are reported to expose degradation at the temporal envelope extremes.

3. Behavioural Quality

Path Tortuosity.

$$\mathcal{T}^{(m)} = \frac{L^{(m)}}{\left\| \mathbf{p}_0^{(m)} - \mathbf{p}^* \right\|_2} \tag{24}$$

where $\mathbf{p}_0^{(m)}$ is the spawn position and $\mathbf{p}^*$ the runway threshold; $\mathcal{T}^{(m)} = 1$ indicates a straight-line trajectory.

Spatial Visitation Entropy. Flight paths are binned into a $B \times B$ histogram, yielding $p_b = \frac{h_b}{\sum_{b'} h_{b'}}$. Shannon entropy [28] over this distribution is:

$$\mathcal{H} = -\sum_{b=1}^{B^2} p_b \log_2 p_b \quad \text{[bits]} \tag{25}$$

Higher $\mathcal{H}$ indicates broader spatial coverage; lower $\mathcal{H}$ indicates route concentration or generalisation failure.

KL Divergence Between Visitation Distributions. The structural routing difference between two configurations A and B is [29]:

$$D_{\text{KL}}\left(P^{(A)} \parallel P^{(B)}\right) = \sum_{b=1}^{B^2} p_b^{(A)} \log_2 \frac{p_b^{(A)}}{p_b^{(B)}} \tag{26}$$

with a small smoothing constant for unvisited bins. Jointly with $\mathcal{H}$ and $\mathcal{T}$, D_{KL} characterises reward topology across geographic sectors.

Failure Mode Analysis. Failed trajectories are characterised by normalised episode progress at termination:

$$\xi^{(m)} = \frac{k^{(m)}}{k_{\max}} \tag{27}$$

where $k^{(m)}$ is the termination step and $k_{\max} = \max_m k^{(m)}$. Failure rates are reported as a function of spawn bearing ψ and, for 4D configurations, as a joint (ψ, τ^*) heatmap to expose systematic generalisation blind spots.

4. Multi-Objective Evaluation and Model Selection Phases

Owing to the two-stage architecture (subsection III.C), model selection is executed sequentially to preserve data dependencies [4].

Phase I: Spatial Policy Selection. Candidate models are evaluated on S_g, $L^{(m)}$, $\mathcal{N}^{(m)}$ and $\mathcal{T}^{(m)}$ via Pareto-efficiency analysis. Selecting and freezing the spatial core is a structural prerequisite: its rollouts must remain invariant to construct the baseline dataset for the DTG sampler (subsubsection III.C.2).

Phase II: Spatial-Temporal Policy Selection. With the spatial core frozen, 4D configurations are evaluated on $\left|\epsilon_{\text{RTA}}^{(m)}\right|$ and composite success rate S. Where configurations are comparable, secondary selection uses $\mathcal{H}$, CV_{tail} and AUC, ensuring convergence stability before deploying the combined policy within the CPS manager.

IV. Experiment Setup

This section details the simulation environment, airspace configuration, agent specification and experimental conditions used to evaluate the GCHRL framework of section III. Deviations from the dual-runway baseline of Groot et al. [4], particularly the airspace scope, are stated explicitly.

A. Simulation Environment

Experiments use Python ($\geq$ 3.10) with BlueSky-gym [15] (Gymnasium-compatible [30], the open-source BlueSky ATM simulator [16] ($\geq$ 1.0.7)). The 4D-Worker is trained with SAC from Stable-Baselines3 ($\geq$ 2.7.1) [31], using `HerReplayBuffer` for HER conditions (subsection II.C). The DTG samplers (subsubsection III.C.2) and the ET surrogate model uses scikit-learn ($\geq$ 1.7.2) [32]. Simulation timestep $\Delta t_{\text{sim}} = 5\,\text{s}$; the agent issues a new heading command every 120 s (24 steps/action).

B. Airspace and Traffic Scenario

While Groot et al. [4] evaluate a simplified two-runway configuration (18R/27), this work trains and evaluates the 4D-Worker across all twelve runway ends of Amsterdam Schiphol (18C, 36C, 18L, 36R, 18R, 36L, 06, 24, 09, 27, 04, 22), each defining an independent goal and exposing the Manager to the full multi-runway TBALP. For direct comparison with Groot et al., final evaluation isolates the 18R/27 configurations to verify whether runway overload is mitigated.

For each runway end, the FAF is placed 25 km along the extended centreline, with the IAF sector 30 km further out, spanning a 60° arc centred on the reverse runway track. The goal vector encodes the assigned runway's IAF position and the RTA component τ^* (subsection II.C). A SINK region bounds each IAF arc; a RESTRICT region penalises premature entry into another runway's funnel.

At reset, a single aircraft spawns at a uniformly sampled bearing around Schiphol, at distance $\sim U(59.5, 285)$ km (FAF+IAF+4.5 km margin to 0.95 $D_{\max}$), heading toward Schiphol. Episodes truncate beyond 315 km (1.05 $D_{\max}$) or $T_{\max} = 21,600$ s (6h, a normalisation horizon only).

C. Aircraft Configuration

A single A320 with default OpenAP [33] performance (as in BlueSky) is used throughout, initialised at 3,000 m and constant CAS 125 m/s; speed/altitude are not control variables, so path-stretching is purely lateral. RECAT-EU wake categories (subsection II.B) are required by the CPS Manager but not reflected in this homogeneous-performance training environment; this gap is treated as an external constraint on the Worker and discussed in subsection III.B.

D. Observation, Action and Reward Specification

The environment exposes a `gymnasium_robotics`-style `Dict` space (`observation`, `achieved_goal`, `desired_goal`), normalised to $[-1.5, 1.5]$ using $D_{\max} = 300$ km and $T_{\max}$. The action is a continuous $(\sin\theta, \cos\theta)$ heading command issued every 120 s.

The base `observation` is (x, y, t), following Groot et al. Since the maximum turn rate makes the reachable heading change within 120 s dependent on current heading ψ, (x, y, t) alone is not Markov (most relevant during holding-pattern manoeuvres). A second variant augments the observation to $(x, y, t, \cos\psi, \sin\psi)$ to restore the Markov property. Both variants are evaluated for effects on convergence and RTA attainment.

The dense step reward combines path-length and noise-exposure terms, weighted -0.0025 and -1.0 respectively, per the R_{aug} decomposition (subsection II.C). Reaching the assigned SINK gives a terminal reward of +10; when τ^* is active, this is scaled by $\max\left(0, 1 - \left(\frac{|\epsilon_{\text{RTA}}|}{\delta_t}\right)^2\right)$, with tolerance $\delta_t = 60\,\text{s}/T_{\max}$ (looser than typical operational tolerances, chosen to remain learnable while still meaningful). This decay applies only when training without HER, since HER assumes $\epsilon_{\text{RTA}} = 0$. Entering a RESTRICT region, reaching an unassigned non-overlapping SINK, or truncation incurs -1. `compute_reward` (used for HER relabelling) omits the RTA term, as relabelled transitions have $\tau^* = t_{\text{achieved}}$ by construction.

E. Training Configuration

Both training stages of subsection III.C share the SAC/HER hyperparameters of Table 2. Policy performance is evaluated at fixed intervals over held-out episodes from the full goal distribution, yielding the curves used for the training-efficiency metrics (subsubsection III.D.1).

Table 2 SAC and HER hyperparameters.

Parameter	Value	Parameter	Value
Network architecture	3 layers, 256 units	Entropy temperature α	Automatic adjustment
Learning rate (actor / critic)	0.0003	HER relabelling strategy	final ($k = 4$)
Discount factor γ	0.99	Training steps (Stage 1, spatial)	250,000
Replay buffer size	10^6	Training steps (Stage 2, spatial-temporal)	350,000
Batch size	256	Random seeds	Env default (unseeded)
Target network update rate (τ_{soft})	0.005	Evaluation cadence	every 5,000 steps, 500 episodes

F. DTG Sampler Dataset

Following subsubsection III.C.1, the trained Stage 1 spatial policy is rolled out until $100,000$ successful episodes are obtained, forming the dataset for fitting the DTG sampler (subsubsection III.C.2); failed episodes are retained for failure-mode analysis (subsubsection III.D.3). As this count depends on each condition's success rate, it is reported per condition. The same rollout set is reused as the comparison baseline for spatial vs. spatial-temporal behavioural differences (subsubsection III.D.3), ensuring observed differences are attributable to the extension rather than to differing samples.

The candidate pool for subsubsection III.C.2 comprises twelve regressors across four families: linear (OLS, Ridge, Lasso, Bayesian Ridge), instance-based (KNN, Radius Neighbours, KDE-based), tree ensembles (Random Forest, Extra Trees, Gradient Boosting, HistGradientBoosting) and MLP. All candidates are screened at default scikit-learn hyperparameters, independently for each combination of coordinate system (Cartesian/polar) and DTG formulation ($\hat{d}_s/\hat{d}_{st}$, Equation 16).

G. Experimental Conditions

Table 3 summarises the conditions, organised per the two-phase selection procedure of subsubsection III.D.4: Phase I selects the frozen spatial policy for DTG sampler construction; Phase II activates the full 4D problem and CPS Manager; Phase III covers CPS operational performance with the Phase II Worker fixed (subsection IV.H). All conditions share Table 2 except where the condition itself is the variable under test. The unconstrained, no-CPS configuration exhibiting Runway Overload is not re-run; results from Groot et al. [4] for this configuration serve as the reference point for CPS-coordinated conditions.

Table 3 Experimental conditions.

Phase	Condition	Purpose
I	Spatial, no HER	Baseline spatial policy
I	Spatial, with HER	Isolates the effect of HER on spatial goal attainment
II	Spatial-temporal, no HER	Full 4D problem without goal re-labelling
II	Spatial-temporal, with HER	Primary 4D-Worker configuration
II	Reward topology variants	Path-length / noise weight sweep, evaluated via spatial visitation entropy and tortuosity
III	CPS, $k = 0$ (FCFS)	Strict FCFS sequencing
III	CPS, $k =$ [TBD]	Relaxed position shifting
III	CPS, $k =$ [TBD], dynamic TTA updates	System resilience under mid-trajectory goal revision
III	CPS with dynamic runway assignment	Load-balance comparison against static runway assignment

H. Evaluation Protocol and Hardware

Each condition is evaluated over $M = 10,000$ episodes, with spawn positions, runway assignments and RTAs resampled per episode (subsection IV.B). Evaluation and behavioural-quality metrics follow subsection III.D; throughput, separation compliance and resilience metrics follow subsubsection III.B.4. All experiments run on a single Apple M3 MacBook Air (8 GB unified memory); batch sizes and parallel environment counts (Table 2) are chosen to fit within this constraint.

V. Results (Deliverable for the Green-Light meeting at 27 weeks)

Make sure that any conclusion/direct answering of the research question we have here, we move to the conclusion or discussion!

Results are presented in the order of the two-phase policy selection introduced in subsubsection III.D.4, followed by a Phase III evaluation of CPS coordination. subsection V.A reports the DTG sampler selection. subsection V.B addresses RQ1.2 and RQ1.3 through the spatial HER ablation and reward-topology analysis. subsection V.C addresses RQ1 and RQ1.1 through the spatial-temporal extension and RTA attainment. subsection V.D addresses RQ2, RQ2.1 and RQ2.2 through CPS throughput, load balance and resilience.

A. DTG Sampler Selection

Runway-by-runway variance metrics, correlation structures, model comparison fields and significance tests are reported in the appendix (paragraph VII). Feature screening revealed that the radial distance feature (r) heavily dominates spatial tracking metrics, whereas Cartesian coordinate offsets (x, y) and angular tracking (θ) exhibit negligible correlations with remaining flight distance. Conditioning on normalised elapsed time (t) yielded a marginal temporal boost of only 1.2–1.3 percentage points over the spatial-only baseline variance reduction of 77.5–77.9%. The polar, spatial-only formulation ($\hat{d}_s$) is selected for production tracking. Cross-validation across ten machine learning architectures confirmed that the Extra Trees regressor achieves a baseline accuracy of $R^2 \geq 0.9994$, with the lowest absolute global errors across all experimental conditions, verified by pairwise Diebold-Mariano and Wilcoxon signed-rank testing ($p < 0.05$ against every alternative).

B. Phase I: Spatial Policy Selection

This section reports convergence efficiency, tail stability and routing characteristics for the four Phase I spatial conditions.

1. HER vs No-HER (RQ1.2)

Figure 2 shows the training curves for all four spatial conditions; Table 4 reports the derived efficiency metrics.

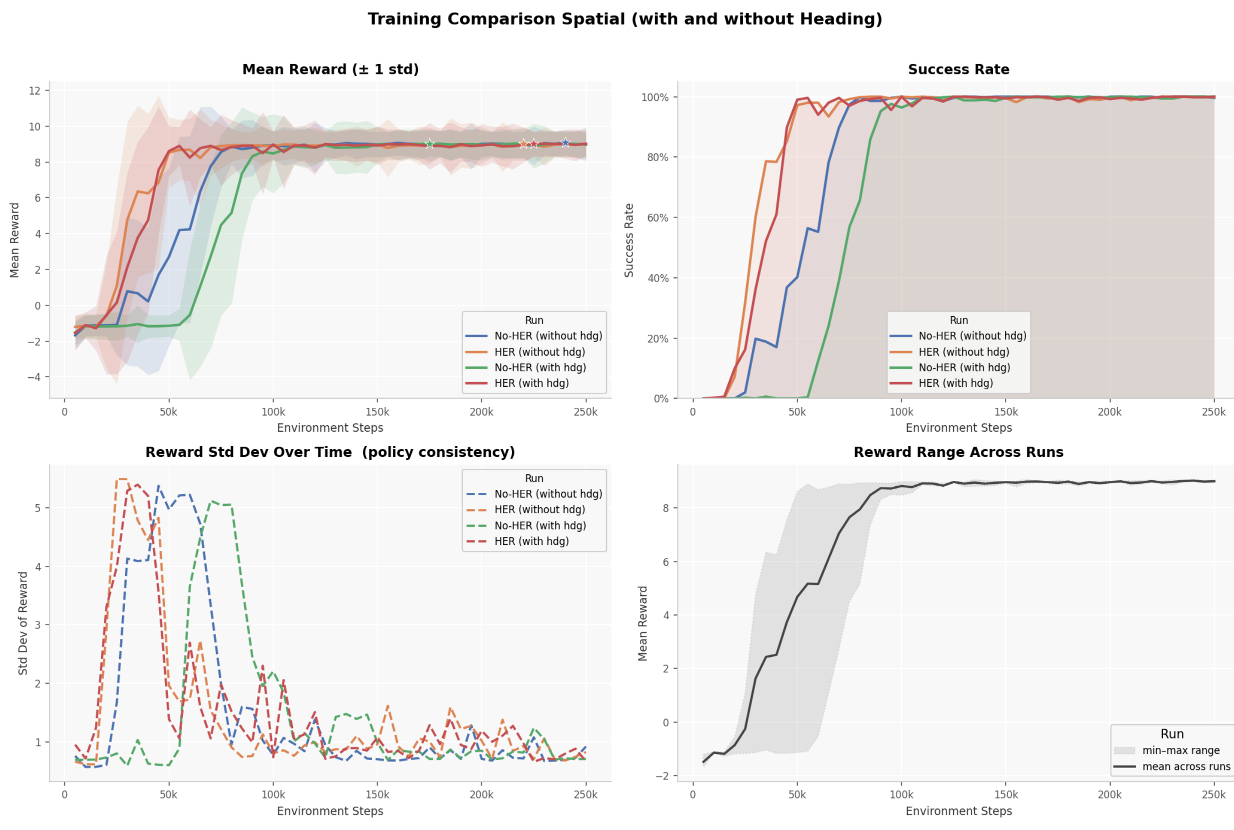


Fig. 2 Mean evaluation reward ($\pm 1\,\sigma$), success rate, reward standard deviation and cross-run reward range during Stage 1 spatial training, for all four observation–relabelling combinations. Stars mark the best checkpoint per run.

HER with $(x, y, t, \cos\psi, \sin\psi)$ reaches $t_{0.80} = 45$ k steps versus 85 k for its no-HER counterpart; the non-augmented pair shows the same ordering (50 k vs 70 k). Δ_{pf} ranges 0.002 (HER, heading) to 0.077 (No-HER, (x, y, t)) and CV_{tail}

Table 4 Training efficiency and asymptotic spatial success rate, HER vs no-HER across observation variants. $t_{0.80}$/$t_{0.90}$: environment steps to 80%/90% of peak success rate; AUC: area under the mean-reward curve (normalised); $\mathrm{CV_{tail}}$: tail coefficient of variation ($\tau = 0.8$); Δ_{pf}: peak-to-final reward drop; S_g: tail-mean success rate.

Condition	$t_{0.80}$ (k steps)	$t_{0.90}$ (k steps)	AUC	$\mathrm{CV_{tail}}$	Δ_{pf}	S_g (%)
No-HER, $(x, y, t, \cos\psi, \sin\psi)$	85	90	6.06	0.00435	0.073	99.84
No-HER, (x, y, t)	70	75	6.96	0.00525	0.077	99.88
HER, $(x, y, t, \cos\psi, \sin\psi)$	45	50	7.64	0.00563	0.002	99.72
HER, (x, y, t)	50	50	7.80	0.00649	0.022	99.74

ranges 0.00435 to 0.00649 across the four conditions (Table 4). All four converge to $S_g \in [99.72\%, 99.88\%]$.

Per-runway diagnostics (Figure 10–Figure 13) show HER with heading achieving $S_{36L} = 100\%$ against $S_{36L} = 94.1\%$ for HER without heading; both No-HER variants sit at 99.8–99.9% on 36L. $T_f \approx 22{-}23$ min and $L \approx 200{-}240$ km are invariant across conditions; reward distributions are concentrated in the 8–10 range, with failure outliers more frequent in No-HER conditions.

2. Reward Topology (RQ1.3)

To calculate visitation entropy and population exposure, agent positions are mapped onto a 2D spatial histogram. Instead of utilising an arbitrary fixed grid, the bin sizes are determined dynamically along each axis using the Freedman–Diaconis rule [34]. This method automatically calibrates the resolution based on both the geographic spread of the trajectories and the total number of data points.

Table 5 Visitation entropy $\mathcal{H}$, tortuosity $\mathcal{T}^{(m)}$, population exposure $\mathcal{N}^{(m)}$ (left) and pairwise KL divergence with derived deltas (right, $\Delta = A - B$).

Condition	$\mathcal{H}$	$\mathcal{T}^{(m)}$	$\mathcal{N}^{(m)}$	Comparison (A – B)	D_{KL}	$\Delta\mathcal{T}^{(m)}$	$\Delta\mathcal{N}^{(m)}$	$\Delta\mathcal{H}$
No-HER, (x, y, t)	14.30	1.149	153.5	No-HER (x, y, t) – HER (x, y, t)	0.10	−0.004	−5.5	+0.06
No-HER, $(x, y, t, \cos\psi, \sin\psi)$	14.35	1.152	152.6	No-HER $(x, y, t, \cos\psi, \sin\psi)$ – HER $(x, y, t, \cos\psi, \sin\psi)$	0.14	−0.027	−0.7	+0.02
HER, (x, y, t)	14.36	1.145	148.0	No-HER (x, y, t) – No-HER $(x, y, t, \cos\psi, \sin\psi)$	0.12	+0.003	−0.9	+0.05
HER, $(x, y, t, \cos\psi, \sin\psi)$	14.34	1.125	151.9	HER (x, y, t) – HER $(x, y, t, \cos\psi, \sin\psi)$	0.15	+0.019	+3.9	+0.02

$\mathcal{H} \in [14.30, 14.36]$ bits and $\mathcal{T}^{(m)} \in [1.125, 1.152]$ across all four conditions (Table 5), with HER $(x, y, t, \cos\psi, \sin\psi)$ lowest on tortuosity and No-HER $(x, y, t, \cos\psi, \sin\psi)$ highest. $\mathcal{N}^{(m)}$ ranges 148.0 (HER, (x, y, t)) to 153.5 (No-HER, (x, y, t)). Pairwise $D_{\mathrm{KL}} \geq 0.10$ for all four comparisons, largest between the two HER conditions (0.15). Within HER, heading augmentation reduces tortuosity ($\Delta\mathcal{T}^{(m)} = -0.027$) without changing entropy; the same comparison without HER gives $D_{\mathrm{KL}} = 0.10$, $\Delta\mathcal{T}^{(m)} = -0.004$.

3. Phase I Selection

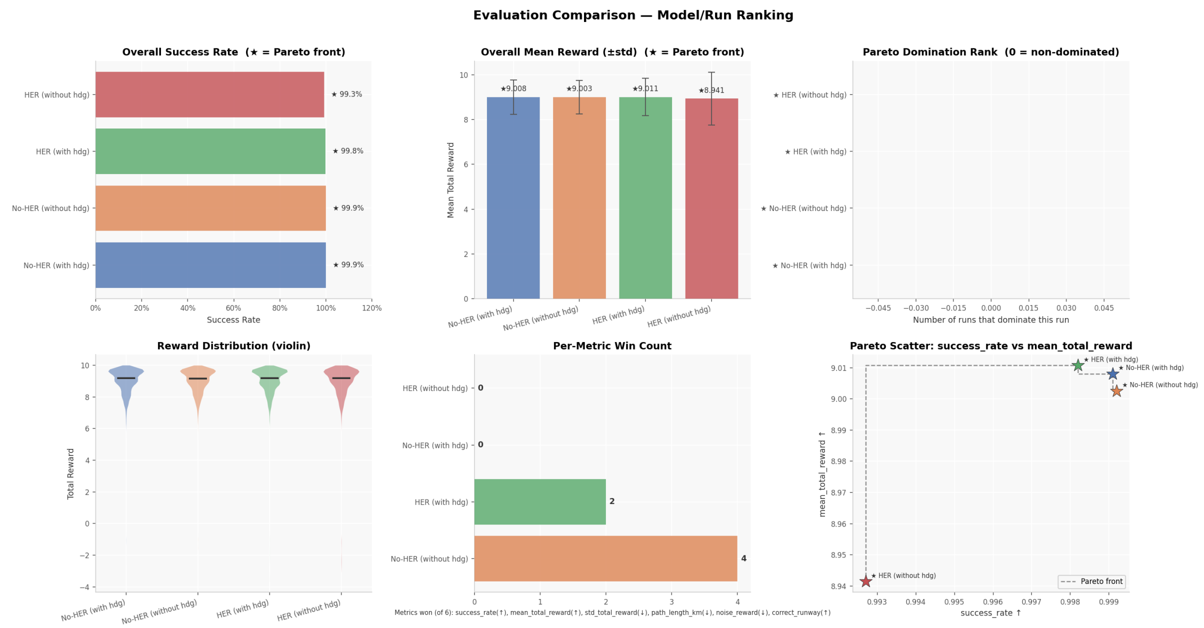


Fig. 3 Pareto comparison across spatial conditions on S_g and mean total reward. All four conditions are non-dominated (rank 0); per-metric win counts and reward distributions included for tiebreaking. Stars denote Pareto-front membership.

All four spatial conditions lie on the non-dominated Pareto front (Figure 3). Selection for Phase II follows a two-criterion cascade: (1) no runway below 99.5% success, (2) absence of generalisation failures. HER with $(x, y, t, \cos\psi, \sin\psi)$ is selected within the with-HER category: all twelve runways achieve $S \geq 99.5\%$ and $\Delta_{pf} = 0.002$. No-HER without heading is selected within the no-HER category: no-HER with heading shows scattered per-runway failures and the same 36L failure mode without HER's compensatory effect (Figure 10–Figure 13). HER with $(x, y, t, \cos\psi, \sin\psi)$ and no-HER without heading are frozen as the spatial cores for Phase II.

C. Phase II: Spatial-Temporal Policy Selection (RQ1, RQ1.1)

1. HER vs No-HER, Spatial-Temporal (RQ1.1)

Stage 2 training extends the frozen Phase I spatial cores with temporal goal injection. Figure 4 and Table 6 report training curves and efficiency metrics for the two observation variants under HER and no-HER; full per-runway diagnostics for all four configurations are reported in the appendix (Figure 14–Figure 17).

No-HER, heading-augmented collapses to $S_g = 74.23\%$ against $S_g \in [97.80, 99.20]\%$ for the remaining three; HER, heading-augmented attains $S_g = 98.86\%$ under the identical observation space (Table 6). Convergence speed: $t_{0.80} = 190$ k (HER, no heading), 215 k (HER, heading), 240 k (No-HER, heading), 265 k (No-HER, no heading). Δ_{pf} and $\mathrm{CV}_{\mathrm{tail}}$ are highest for No-HER, heading-augmented (0.446, 0.064) and lowest for HER, heading-augmented (0.079,

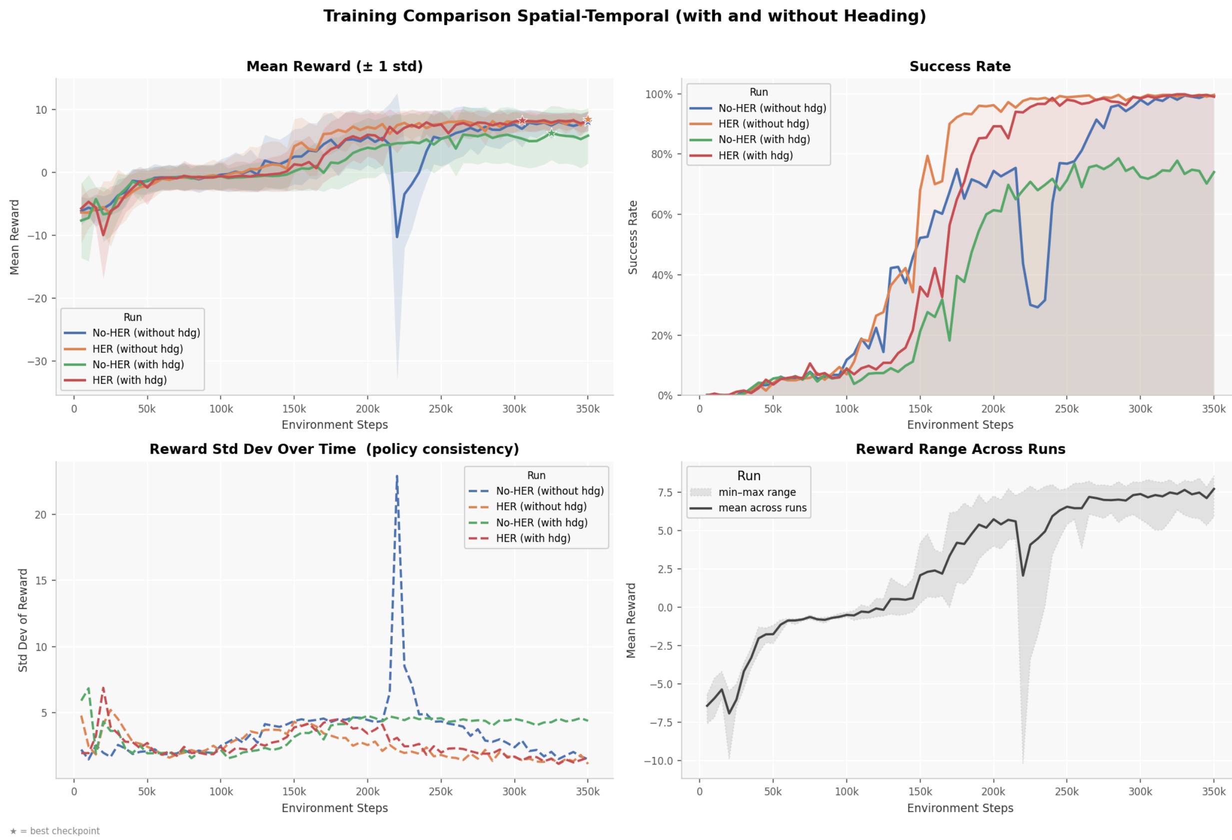


Fig. 4 Mean evaluation reward (±1 σ), success rate, reward standard deviation and cross-run reward range during Stage 2 spatial-temporal training, for all four observation–relabelling combinations. Stars mark the best checkpoint per run.

0.031).

Flight time and path length expand under RTA pressure relative to Phase I: $T_f \approx 32{-}38$ min (vs 22–23 min) and $L \approx 200{-}400$ km (vs 200–240 km). On-time fraction ranges 0.9–1.0 across HER conditions. $S_g \in [99.1\%, 99.3\%]$ for HER conditions and 99.4% for No-HER without heading, with correct-runway assignment $\approx$ 1.0 across all twelve runways; No-HER with heading instead collapses on 36L/36C/36R (S = 79.6%/60.8%/76.9%, correct-runway assignment 0.2–0.3).

2. RTA Attainment and Arrival Delay (RQ1.1)

Figure 5 isolates the effect of the heading objective on arrival-delay precision, decomposing the on-time rates summarised above.

Without heading: on-time rate 99.6%/99.5% (no-HER/HER), MAE 0.18/0.16 min, with negligible clipping ($\leq$ 0.004%) given a tail extending to 56.5/38.7 min (Figure 5). Both distributions are strongly leptokurtic (K = 2075.34/1479.06, S = 38.58/35.74): on-time mass dominates, but rare large outliers inflate the tail statistics far beyond the visible ±1 min band. With heading: no-HER on-time rate drops to 75.7%, MAE rises to 4.36 min, and the KDE

Table 6 Training efficiency and asymptotic spatio-temporal success rate, HER vs no-HER across observation variants. $t_{0.80}$**/**$t_{0.90}$**: environment steps to 80%/90% of peak success rate; AUC: area under the mean-reward curve (normalised); CV**$_{\text{tail}}$**: tail coefficient of variation (**$\tau = 0.8$**);** Δ_{pf}**: peak-to-final reward drop;** S_g**: tail-mean success rate.**

Condition	$t_{0.80}$ (k steps)	$t_{0.90}$ (k steps)	AUC	CV$_{\text{tail}}$	Δ_{pf}	S_g (%)
No-HER, $(x, y, t, \cos\psi, \sin\psi)$	240	255	1.781	0.064	0.446	74.23
No-HER, (x, y, t)	265	295	2.402	0.056	0.000	97.80
HER, $(x, y, t, \cos\psi, \sin\psi)$	215	230	2.994	0.031	0.079	98.86
HER, (x, y, t)	190	210	3.553	0.041	0.000	99.20

bandwidth widens an order of magnitude ($h = 0.85$ vs 0.07); skewness and kurtosis fall to 2.16 and 3.74, still flagged but close to normal, indicating a systematic spread of delay rather than rare outliers, with a secondary KDE mode around 10–20 min consistent with the 36L/36C/36R runway failure mode (subsubsection V.C.5). HER with heading on-time rate remains 99.6%, MAE 0.20 min, with intermediate tail heaviness ($S = 27.14$, $K = 814.48$) between the no-heading and no-HER-with-heading conditions.

3. Behavioural Quality: Spatial vs Spatial-Temporal

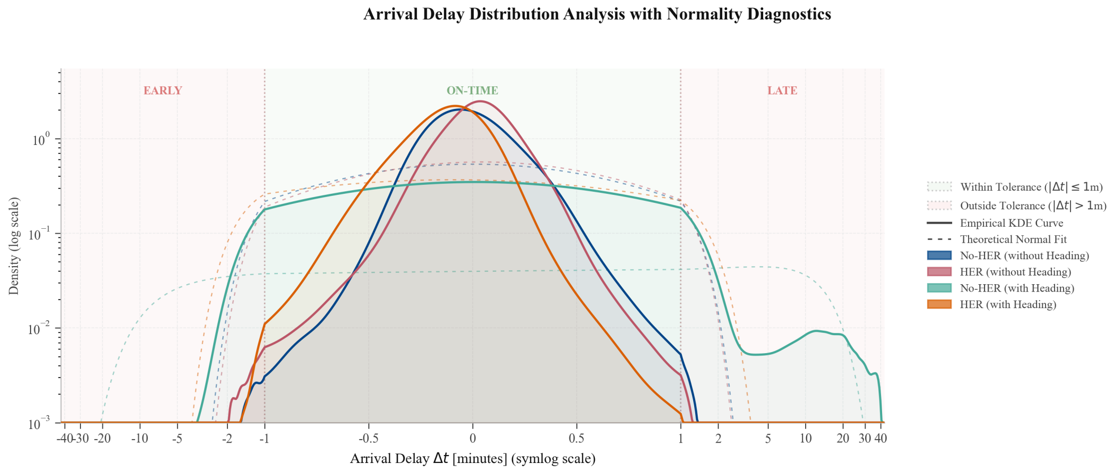
This subsubsection compares trajectory-level behavioural metrics, path length $L^{(m)}$, population exposure $\mathcal{N}^{(m)}$, tortuosity $\mathcal{T}^{(m)}$ and visitation entropy $\mathcal{H}$, between the Phase II spatio-temporal conditions and the frozen spatial core on the shared rollout dataset, isolating the effect of temporal goal injection.

Table 7 Visitation entropy $\mathcal{H}$**, tortuosity** $\mathcal{T}^{(m)}$**, population exposure** $\mathcal{N}^{(m)}$ **(left) and pairwise KL divergence with derived deltas (right,** $\Delta = A - B$**).**

Condition	$\mathcal{H}$	$\mathcal{T}^{(m)}$	$\mathcal{N}^{(m)}$	Comparison (A – B)	D_{KL}	$\Delta\mathcal{T}^{(m)}$	$\Delta\mathcal{N}^{(m)}$	$\Delta\mathcal{H}$
No-HER, (x, y, t)	14.26	1.669	144.6	No-HER (x, y, t) – HER (x, y, t)	0.23	−0.007	−0.3	+0.01
No-HER, $(x, y, t, \cos\psi, \sin\psi)$	14.19	1.679	102.7	No-HER $(x, y, t, \cos\psi, \sin\psi)$ – HER $(x, y, t, \cos\psi, \sin\psi)$	0.41	−0.013	−49.9	−0.10
HER, (x, y, t)	14.25	1.676	144.9	No-HER (x, y, t) – No-HER $(x, y, t, \cos\psi, \sin\psi)$	0.46	−0.10	+41.9	+0.073
HER, $(x, y, t, \cos\psi, \sin\psi)$	14.35	1.692	152.5	HER (x, y, t) – HER $(x, y, t, \cos\psi, \sin\psi)$	0.26	−0.016	−7.6	−0.105

$\mathcal{H} \in [14.19, 14.35]$ bits and $\mathcal{T}^{(m)} \in [1.669, 1.692]$ across the four conditions (Table 7), versus $[1.125, 1.152]$ in Phase I. No-HER, $(x, y, t, \cos\psi, \sin\psi)$ has the lowest $\mathcal{N}^{(m)}$ (102.7) against 144.6–152.5 for the other three. Pairwise D_{KL} ranges 0.23–0.46, versus 0.10–0.15 in Phase I, largest between the two No-HER conditions (0.46).

Figure 6 compares the selected Phase II configuration, No-HER (x, y, t), against its frozen spatial-only core: $\mathcal{T}^{(m)} = 1.669$ vs 1.149 ($\Delta = +0.520$), $\mathcal{N}^{(m)} = 144.6$ vs 153.5 ($\Delta = -8.9$), $\mathcal{H} = 14.26$ vs 14.42 bits ($\Delta = -0.165$),



Dataset Source	N	On-Time Rate	Delayed Rate	MAE	Max Abs Delay	Clipped (%)	KDE BW (h)	Skewness (S)	Ex. Kurtosis (K)
No-HER (without Heading)	100,510	99.6%	0.4%	0.18 min	56.5 min	0.004%	0.07	**38.58**	**2075.34**
HER (without Heading)	100,610	99.5%	0.5%	0.16 min	38.7 min	0.000%	0.07	**35.74**	**1479.06**
No-HER (with Heading)	132,309	75.7%	24.3%	4.36 min	45.7 min	0.010%	0.85	**2.16**	**3.74**
HER (with Heading)	100,834	99.6%	0.4%	0.20 min	44.3 min	0.001%	0.11	**27.14**	**814.48**

Bold red: $|S| > 1$ or $|K| > 3$ (deviation from normality).

Fig. 5 Distribution of arrival delay Δt for HER and No-HER agents, with and without the heading objective, on a symlog axis with the ± 60 s on-time tolerance band shaded; each empirical KDE curve is overlaid with its theoretical normal fit (dashed). Per-condition sample size, on-time/delayed rate, MAE, maximum absolute delay, clipped fraction, KDE bandwidth, skewness (S) and excess kurtosis (K) are summarised below; bold red flags $|S| > 1$ or $|K| > 3$.

$D_{KL} = 0.60$.* The difference panel's $\Delta h > 0$ region (dashed orange) forms a ring north, east and south of EHAM; the $\Delta h < 0$ region (dashed teal) sits to the south and southwest; the nested grey envelopes (80%, 92%, 99% of cumulative $|\Delta h|$) are concentrated within roughly 150 km of the field.

4. Phase II Selection

Update text to make No-HER without Heading the winner!

Selection uses success rate S and RTA error $|\epsilon_{\mathrm{RTA}}^{(m)}|$ first, then $\mathcal{H}$, $\mathrm{CV}_{\mathrm{tail}}$ and AUC as tiebreakers. Figure 7: three of the four configurations are non-dominated (rank 0); HER, heading-augmented is dominated (rank 1) on S_g and mean total reward.

The selection cascade is: (1) per-runway $S > 99\%$, no runway below 98.5%; (2) $\mathcal{P}(\text{On Time}) \geq 0.9$; (3) Correct Runway ≥ 0.99. No-HER with heading is excluded on criterion (1): $S \in [60.8\%, 79.6\%]$ on 36L/36C/36R. The remaining three candidates all satisfy criteria (1)–(3); HER with heading is Pareto-dominated on RTA accuracy (0.95 vs

*Both panels in this figure share a single bin width per axis, fixed via the Freedman–Diaconis rule (subsubsection V.B.2) on the spatio-temporal rollout, which has higher dispersion than the spatial-only rollout. Reusing this finer, spatio-temporal-derived grid for the spatial-only panel raises its entropy slightly relative to the value computed with its own, independently-tuned bin width in Table 5 (14.30 bits).

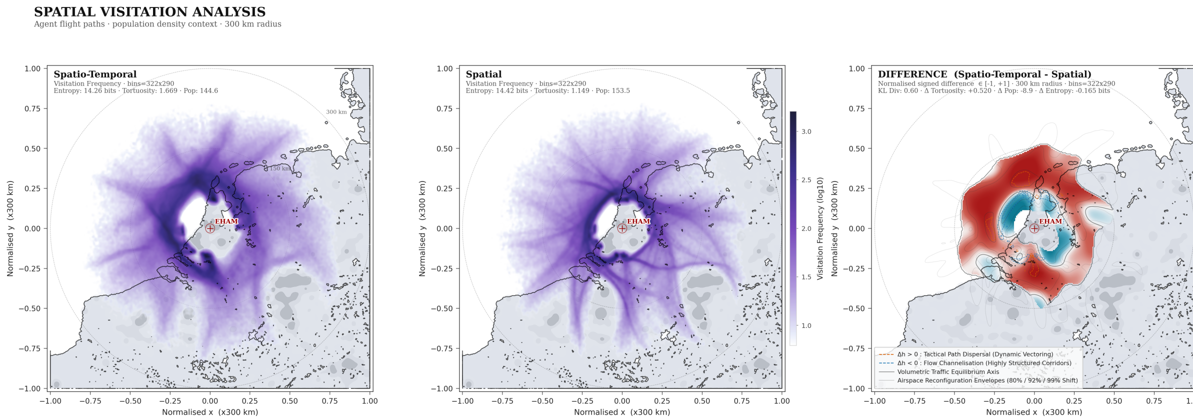


Fig. 6 Spatial visitation density for the Phase II winner, No-HER (x, y, t): spatio-temporal policy (left) vs its frozen spatial-only core (centre), with the normalised signed difference (right, 300 km radius). Red indicates higher spatio-temporal visitation; teal indicates higher spatial-only visitation.

0.97 mean) and total reward, with $S_g = 99.1\%$ against $S_g = 99.4\%$ for no-HER without heading. No-HER without heading is selected as the Phase II configuration, on lowest RTA error variance and tightest correct-runway distribution across runways.

5. Failure Mode Analysis: No-HER with Heading

No-HER with heading: $S \approx 3.5$–3.8% on 36L/36C/36R versus $> 99\%$ elsewhere (Table 8); correct-runway assignment 0.25–0.27, on-time rate 0.04–0.05. Failure rate as a function of spawn bearing and RTA is ≈ 1 outside a window of bearing 90–150°and RTA 0.02–0.14.

Table 8 Per-runway outcome statistics for No-HER with heading. Path length, RTA, heading error, and tortuosity values represent the mean calculated for successful (S) and failed (F) runs.

Runway (Heading)	n_S	n_F	S (%)	Mean Values (S / F)			
				Path (km)	RTA	Hdg. err. (°)	Tortuosity
36L (2°)	415	10561	3.8	254 / 146	0.082 / 0.095	−10.8 / −1.2	1.42 / 1.09
36C (3°)	396	10592	3.6	256 / 141	0.082 / 0.094	−13.6 / 4.1	1.45 / 1.08
36R (2°)	385	10606	3.5	248 / 141	0.079 / 0.095	−12.5 / 5.1	1.44 / 1.08

Failed runs: path length 141–146 km vs 248–256 km (success), tortuosity 1.08–1.09 vs 1.42–1.45, RTA 0.094–0.095 vs 0.079–0.082 (normalised time units). Heading error: −10.8/−13.6/−12.5 (success) vs −1.2/+4.1/+5.1 (failure) for 36L/36C/36R respectively.

D. Phase III: Hierarchical CPS Coordination (RQ2)

Phase III holds the Phase II 4D-Worker configuration fixed and varies only the CPS coordination layer.

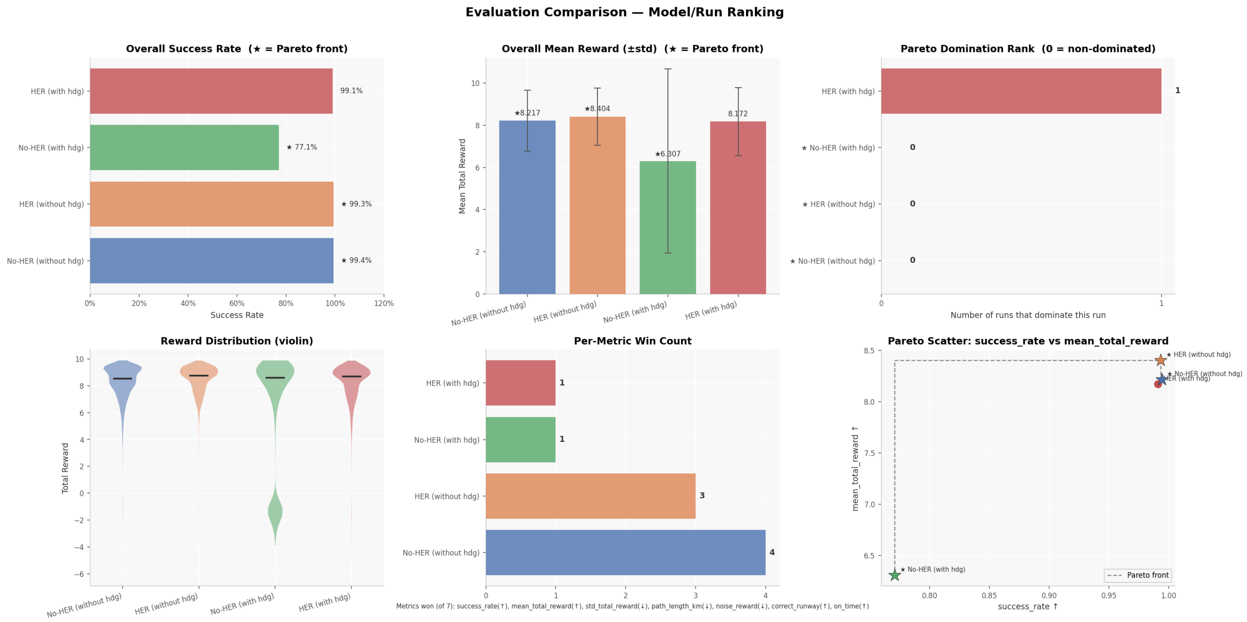


Fig. 7 Pareto comparison across spatio-temporal conditions on S_g and mean total reward. Three conditions are non-dominated (rank 0), only *HER (with hdg)* is dominated (rank 1); per-metric win counts and reward distributions included for tie-breaking. Stars denote Pareto-front membership.

1. Throughput and Separation Compliance (RQ2.1)

This subsubsection reports per-runway throughput Γ_r, total throughput Γ, per-runway separation compliance $C_{\mathrm{sep},r}$ and total separation compliance C_{sep} for CPS conditions $k = 0$ and $k =$ `[TBD]`, benchmarked against the no-CPS baseline of Groot et al. [4].

Table 9 Throughput and separation compliance, CPS conditions vs Groot et al. baseline.

2. Runway Load Balance / Runway Overload (RQ2.1)

This subsubsection compares per-runway throughput on 18R/27 between the CPS conditions and the Groot et al. baseline, and between static and dynamic runway assignment, to assess Runway Overload mitigation.

Fig. 8 Per-runway throughput: Groot et al. baseline vs static and dynamic CPS assignment.

3. Resilience Under Dynamic TTA Updates (RQ2.2)

This subsubsection reports the recovery error $\Delta\epsilon^{(m)}$ and recovery rate R_{rec} for dynamic versus static TTA conditions.

4. Delay Ripple Effect

This subsubsection reports the delay ripple index ρ_{ripple} per CPS condition, indicating whether tracking errors propagate downstream or are damped.

Table 10 Delay ripple index ρ_{ripple} across CPS conditions.

VI. Discussion (Optional deliverable for the Green-Light meeting at 27 weeks)

The Extra Trees regressor selected for the DTG sampler (subsection V.A) has a deployment cost the R^2 comparison doesn't show: the default 100-estimator ensemble needs ≈ 40 GB of memory versus ≈ 1.5 GB at $n_{\text{estimators}} = 15$, with negligible R^2/RMSE loss. Since the sampler runs continuously alongside the RL environment, the reduced configuration is the one to deploy.

HER's Phase I convergence advantage was hypothesised to matter more under Phase II's sparse RTA reward, with non-HER variants failing to converge. This only partly holds: No-HER, (x, y, t) converges fine ($S_g = 97.80\%$); the failure is specific to No-HER plus heading ($S_g = 74.23\%$), which HER alone resolves. HER's contribution under RTA pressure is therefore better read as a robustness margin against the heading liability, not a general convergence safeguard, and its asymptotic edge over No-HER (negligible in Phase I) only appears once RTA is added.

Heading augmentation was introduced to restore the Markov property during holding patterns, motivated by RTA-tracking ambiguity. The data doesn't support this: paired with HER it gives no RTA-accuracy benefit (0.95 vs 0.97 mean) and near-identical on-time rates; paired with no-HER it is actively harmful. The selected Phase III configuration carries no heading state, suggesting the ambiguity is either not a practical cost or is more cheaply resolved by relabelling alone.

Heading-augmented HER has the lowest Phase I tortuosity of all four conditions without any change in entropy, and the convergence-speed gap within the HER pair is too small (45 k vs 50 k) to explain it. This looks like a genuine geometric effect of heading state under relabelled training rather than an artefact of training time: a smoother-path benefit distinct from the convergence speed it was originally framed around.

RTA attainment here relies entirely on lateral path-stretching, since aircraft are modelled as a homogeneous, constant-speed A320 fleet; the holding-pattern and S-curve-dogleg manoeuvres behind the Phase II tortuosity increase (subsubsection V.C.3) are evidence of this constraint, not a free choice. In a mixed-fleet setting, speed control would offer an additional lever for timing, potentially reducing both path length and population exposure relative to the constant-speed case tested here, but would also complicate inter-category separation, which depends on relative speed as well as position.

The 36L/36C/36R failure mode is absent under HER and under no-HER without heading, localising the cause to the heading–RTA interaction rather than RESTRICT or buffer definitions. The inconsistent sign of the heading error across runways rules out a single directional bias; runway misidentification among the closely-spaced parallels under temporal pressure is the more consistent explanation.

The Phase III winner, HER without heading, is not the strongest Pareto candidate: HER with heading is only marginally dominated and satisfies all three selection criteria. The deciding factor is the computational cost of carrying heading state into Phase III's CPS coordination loop, a deliberate trade of marginal RTA accuracy (0.95 vs 0.97) for a cheaper policy, worth revisiting once Phase III throughput is measured.

The DTG sampler's Extra Trees architecture was validated on path-length regression; the ETA surrogate (subsubsection III.B.1) shares the same input rollouts but predicts a different target, and tree-ensemble performance doesn't transfer automatically across target types. Whether Extra Trees is independently validated for the ETA surrogate, or assumed by analogy, should be stated explicitly.

The static-assignment limitation, load imbalance once a dominant IAF sector saturates one runway, is meant to be addressed by the dynamic-assignment condition, but the throughput and load-balance results to substantiate this are not yet populated (Table 9, Figure 8). Complete this once those results are available; demand-anticipating allocation is a natural extension beyond the static/dynamic contrast tested here.

DISCUSSION ITEM: Isolate Scheduling vs. Runway Optimisation Effects

Add a paragraph explicitly clarifying that because our dynamic runway module operates greedily on raw, unconstrained ETAs before sequencing, it has no systemic awareness of queue density or downstream wake conflicts. We need to state clearly that this makes our method a strict lower bound on joint optimisation. Crucially, emphasise that because the runway choice is frozen prior to scheduling, any observed improvements in throughput, separation compliance, or delay mitigation cannot be credited to an over-engineered runway allocation trick. Instead, it isolates the variables perfectly, proving that our positive results are driven entirely by the core contributions: the k-CPS constraints and the tactical path-stretching robustness of our RL worker agent. This frames a simplification as a rigorous scientific control!

DISCUSSION ITEM: Future-Proofing via Distance-to-Go (DTG) Metrics

Add a paragraph explaining the forward-looking architectural rationale for using DTG in the training stage versus time.

Clarify that by predicting DTG, the neural network learns to map its current 2D flight-path geometry directly to a physical remaining distance. When moving from the purely spatial core to the spatio-temporal (4D) worker configuration, this physical distance is what allows the agent to calculate its own internal, feasible arrival time window based on its speed envelope.

Highlight that this deliberately future-proofs the 4D worker agent's policy. Should the environment be extended to include variable speed profiles, acceleration/deceleration constraints, or wind vectors, the absolute arrival times will shift, but the underlying spatial core and its mapping to physical distance remain entirely robust and unchanged.

VII. Conclusion (Deliverable for the Green-Light meeting at 27 weeks)

The DTG sampler is finalised: a polar, spatial-only Extra Trees regressor ($\hat{d}_s$), selected over ten candidate architectures, achieves $R^2 \geq 0.9994$ with the lowest global error, confirmed by Diebold-Mariano and Wilcoxon testing. Radial distance dominates the feature set; angular, Cartesian and explicit temporal features add negligible value.

RQ1.2: HER gives a convergence-rate advantage in Phase I without changing asymptotic spatial success ($S_g \in [99.72\%, 99.88\%]$ across all four conditions). This does not transfer unchanged to Phase II: under the RTA-shaped reward, HER's role shifts from a speed bonus to a robustness requirement, since No-HER with heading collapses ($S_g = 74.23\%$) while every HER condition trains successfully.

RQ1/RQ1.1: spatial-temporal extension with RTA goal injection is learnable, and on-time attainment is achievable for three of the four configurations, but only because HER compensates for the heading liability already visible in Phase I. The configuration carried to Phase III is HER without heading, neither of the two configurations frozen after Phase I selection. Phase II's full re-test of all four combinations was necessary to find it; the Phase I two-track screening is a valid early filter but not a substitute for that re-test.

Phase III (RQ2, RQ2.1, RQ2.2) is not yet complete: throughput, load-balance and resilience results are placeholders pending the CPS condition k. This conclusion should be added once those results are populated.

Appendix

Statistical Tests for DTG Sampler Model Selection

Two paired tests are applied to cross-validated per-sample prediction errors to confirm that performance differences between candidate DTG regressors are not attributable to sampling variability.

Diebold-Mariano Test. The Diebold-Mariano (DM) test [26] evaluates the null hypothesis of equal predictive accuracy between two competing forecast sequences. Let $e_i^{(A)}$ and $e_i^{(B)}$ denote the per-sample errors of models A and B on held-out fold observations, and define the loss differential $d_i = g(e_i^{(A)}) - g(e_i^{(B)})$ for a loss function g (here, squared error). The DM statistic is:

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})/n}} \tag{28}$$

where $\bar{d}$ is the mean loss differential and $\hat{V}(\bar{d})$ is a heteroscedasticity-consistent variance estimate. Following Diebold [35], held-out fold predictions are treated as forecast sequences; AIC/BIC are unsuitable here as several candidate models lack a well-defined likelihood. A positive DM statistic indicates model A yields larger errors than model B; significance is assessed at $p < 0.05$.

Wilcoxon Signed-Rank Test. The Wilcoxon signed-rank test [27] provides a non-parametric complement to the DM test, making no distributional assumption on the error differences d_i. The test ranks the absolute differences $|d_i|$ and evaluates whether the signed ranks are symmetrically distributed around zero under the null of no systematic difference. It guards against inflated DM statistics arising from heavy-tailed or skewed error distributions.

Combined Decision Rule. A performance difference between two candidate models is accepted as statistically significant only when both the DM test and the Wilcoxon test independently agree at $p < 0.05$. This joint criterion reduces the risk of false positives from either parametric assumptions (DM) or low power under near-symmetric distributions (Wilcoxon). Results for all pairwise comparisons against the selected Extra Trees surrogate are reported in Table 13.

DTG Sampler Calibration and Validation Metrics

This section evaluates the DTG sampler configuration. Table 11 reports per-runway variance reduction (spatial, temporal, total); Table 12 gives Pearson correlations between candidate features and distance-to-go; Figure 9 compares the top six regressors across four metrics and HER/heading conditions; Table 13 reports Diebold-Mariano statistics confirming the ET sampler significantly outperforms all baselines.

Per-Configuration Evaluation Dashboards

Figure 10–Figure 17 give the full per-configuration diagnostics underlying subsection V.B and subsection V.C. Each dashboard breaks down one HER/heading/stage combination by runway group (04, 06, 09, 18C/L/R, 22, 24, 27, 36C/L/R): reward distribution (success/failure overlay), success rate with episode count, reward timeline with rolling mean, and flight time, path length and noise reward (mean ± std); spatial-temporal dashboards add correct-runway and on-time rate.

Table 11 DTG variance reduction (%) per runway: spatial-only ($\hat{d}_s$), temporal boost and total ($\hat{d}_{st}$), across heading-observation and HER conditions.

	Without heading						With heading					
	HER			No-HER			HER			No-HER		
Runway	Spatial	Temporal	Total	Spatial	Temporal	Total	Spatial	Temporal	Total	Spatial	Temporal	Total
04	80.04	1.38	81.41	80.33	1.32	81.65	79.22	1.61	80.83	80.38	1.33	81.71
06	78.97	1.27	80.24	79.66	1.49	81.15	78.76	1.21	79.97	79.30	1.13	80.42
09	77.83	1.03	78.87	77.98	1.57	79.56	77.08	0.83	77.91	77.69	0.85	78.53
18C	77.08	1.09	78.17	76.80	1.58	78.38	76.40	0.92	77.33	76.68	0.81	77.49
18L	76.87	1.30	78.18	77.04	1.93	78.97	76.44	0.73	77.17	76.56	0.77	77.33
18R	76.81	0.87	77.68	76.65	1.80	78.45	76.23	1.14	77.37	76.46	0.80	77.27
22	77.24	1.62	78.86	76.83	2.16	78.99	77.13	1.83	78.96	76.79	2.21	79.00
24	77.77	1.98	79.75	77.75	1.23	78.99	77.58	2.07	79.66	77.37	1.90	79.26
27	78.70	0.84	79.54	79.13	0.56	79.68	78.54	1.33	79.87	77.87	0.85	78.72
36C	77.17	1.03	78.20	77.52	0.69	78.20	77.01	1.32	78.33	78.31	1.79	80.11
36L	77.56	0.55	78.11	77.67	0.80	78.47	77.51	0.71	78.22	78.41	1.63	80.04
36R	77.64	1.00	78.64	77.73	0.78	78.51	77.78	1.11	78.89	78.56	0.70	79.26
Mean	77.81	1.16	78.97	77.92	1.33	79.25	77.47	1.23	78.70	77.87	1.23	79.10

Table 12 Pearson correlation r between candidate features and distance-to-go, across configurations. All correlations significant at $p < 0.001$ except where noted. Spearman ρ follows the same pattern and is omitted for brevity.

Feature	HER, no hdg	HER, hdg	No-HER, no hdg	No-HER, hdg
x	+0.044	+0.047	+0.049	+0.054
y	-0.024	-0.040	-0.019	+0.017
r	+0.792	+0.805	+0.783	+0.788
θ	+0.028	$-0.000^{\dagger}$	+0.018	+0.045

$^{\dagger}p = 0.939$, not significant.

Table 13 Diebold-Mariano statistic (last fold) for ETDTGSampler vs. each challenger, across the four HER/heading-observation conditions. All Wilcoxon tests $p < 0.0001$ unless noted. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns: not significant.

Challenger	No-HER	HER	No-HER, hdg	HER, hdg
RF	$+3.07^{**}$	$+0.29^{ns}$	$+4.35^{***}$	$+1.98^{*}$
KNN	$+19.77^{***}$	$+20.91^{***}$	$+24.54^{***}$	$+20.25^{***}$
HistGB	$+19.59^{***}$	$+22.19^{***}$	$+24.90^{***}$	$+21.61^{***}$
MLP	$+43.46^{***}$	$+50.99^{***}$	$+46.31^{***}$	$+40.15^{***}$
GB	$+65.92^{***}$	$+65.24^{***}$	$+61.22^{***}$	$+67.76^{***}$
Ridge/BayesRidge/Linear	$+225.56^{***}$	$+226.16^{***}$	$+215.89^{***}$	$+227.82^{***}$
Lasso	$+223.44^{***}$	$+224.62^{***}$	$+214.08^{***}$	$+225.62^{***}$

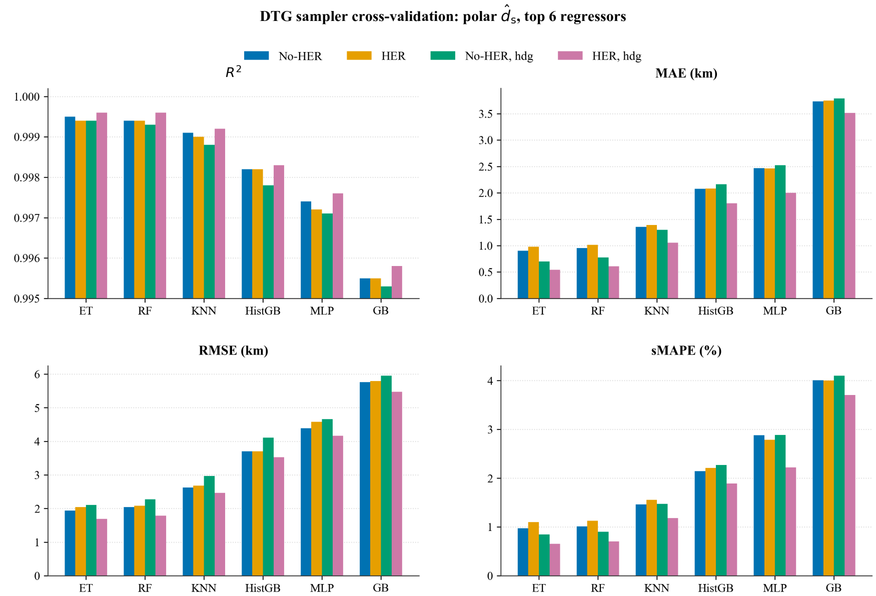


Fig. 9 Cross-validated R^2, MAE, RMSE and sMAPE for the top 6 of 10 candidate regressors, polar $\hat{d}_s$ configuration, across the four HER/heading conditions. The four linear models (Linear, Ridge, BayesRidge, Lasso) are excluded: they cluster at $R^2 \approx 0.68$, MAE ≈ 39–$40\,km$, RMSE ≈ 47–$50\,km$, sMAPE $\approx 36\%$ (see Table 13).

Phase I: Spatial

Phase II: Spatial-Temporal

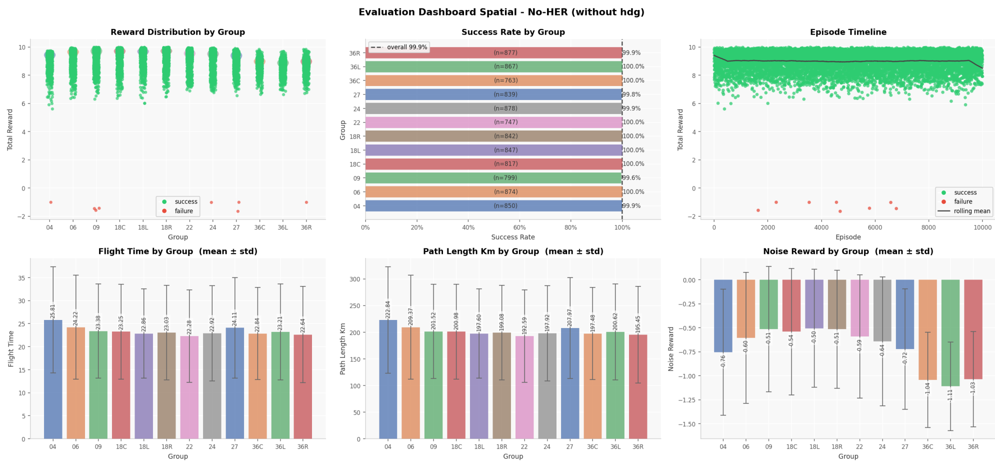


Fig. 10 Per-runway evaluation dashboard, Phase I spatial, No-HER without heading. Panels, left to right, top to bottom: reward distribution by group (success/failure overlay), success rate by group (with episode count n), episode timeline with rolling mean, flight time by group, path length by group, noise reward by group.

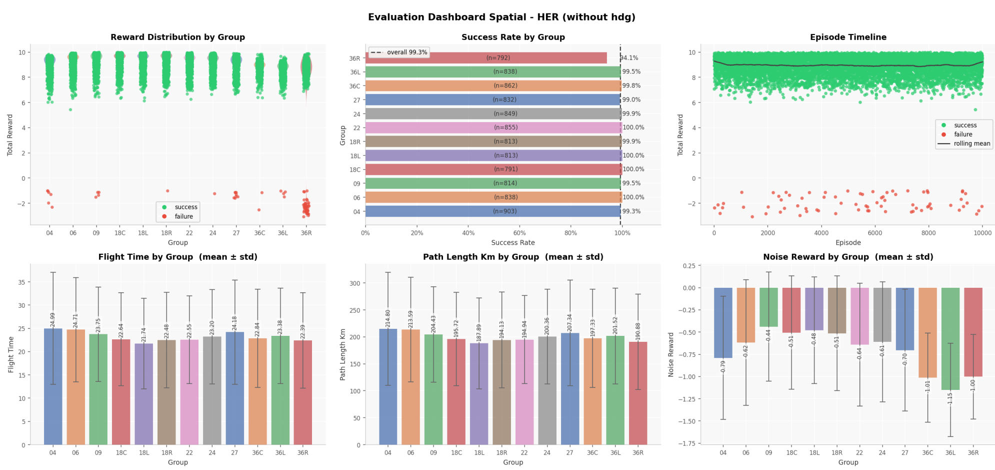


Fig. 11 Per-runway evaluation dashboard, Phase I spatial, HER without heading. Panels same as in Figure 10.

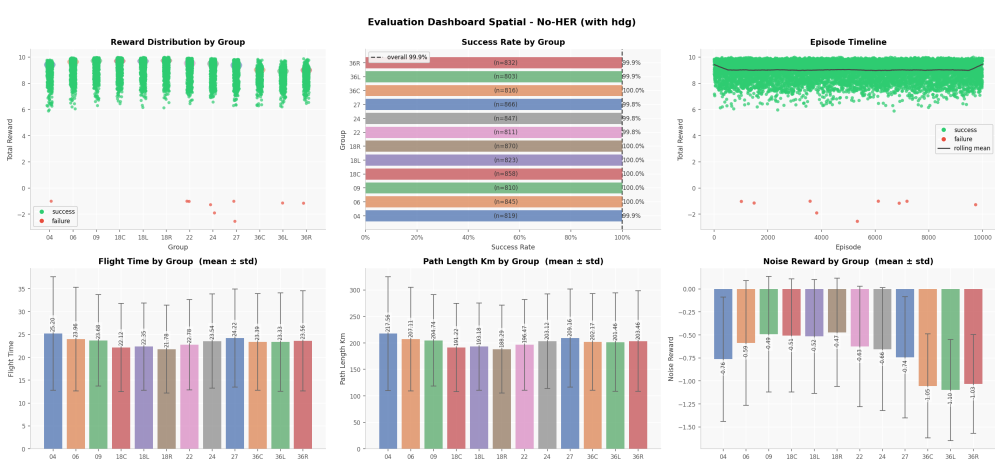


Fig. 12 Per-runway evaluation dashboard, Phase I spatial, No-HER with heading. Panels same as in Figure 10.

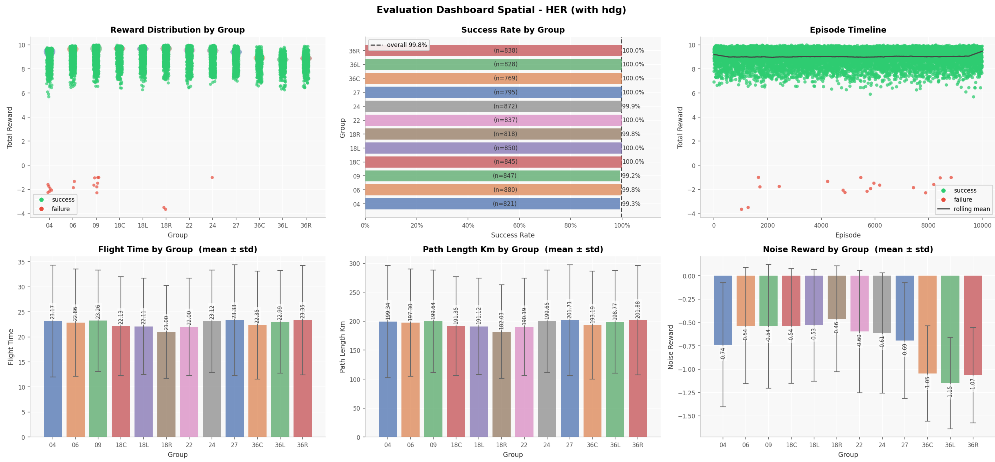


Fig. 13 Per-runway evaluation dashboard, Phase I spatial, HER with heading. Panels same as in Figure 10.

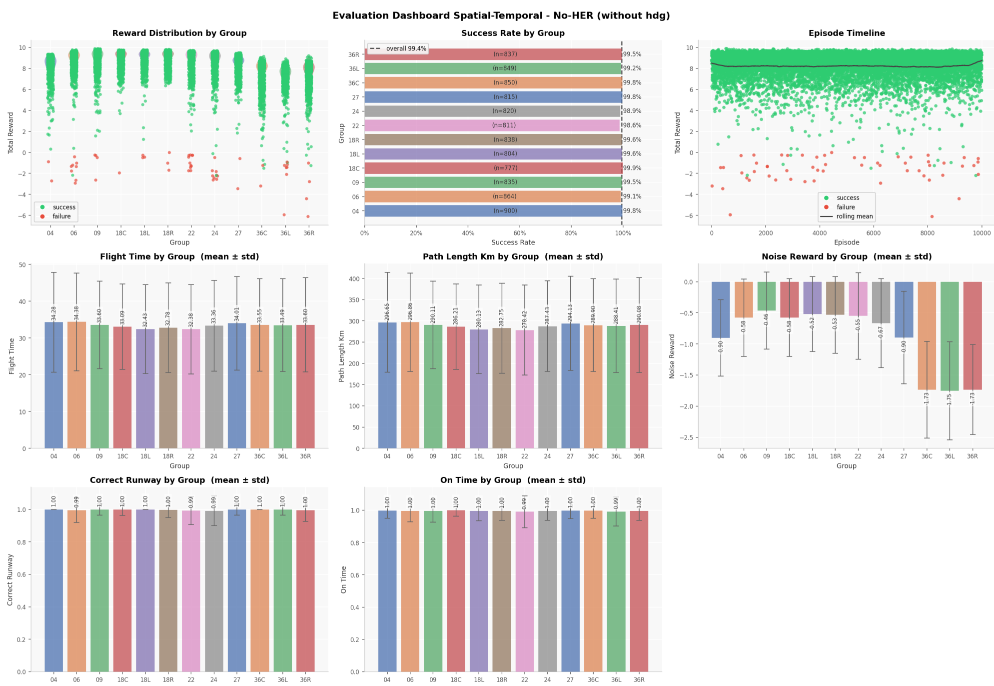


Fig. 14 Per-runway evaluation dashboard, Phase II spatial-temporal, No-HER without heading. Panels same as in Figure 10, plus correct-runway rate and on-time rate by group.

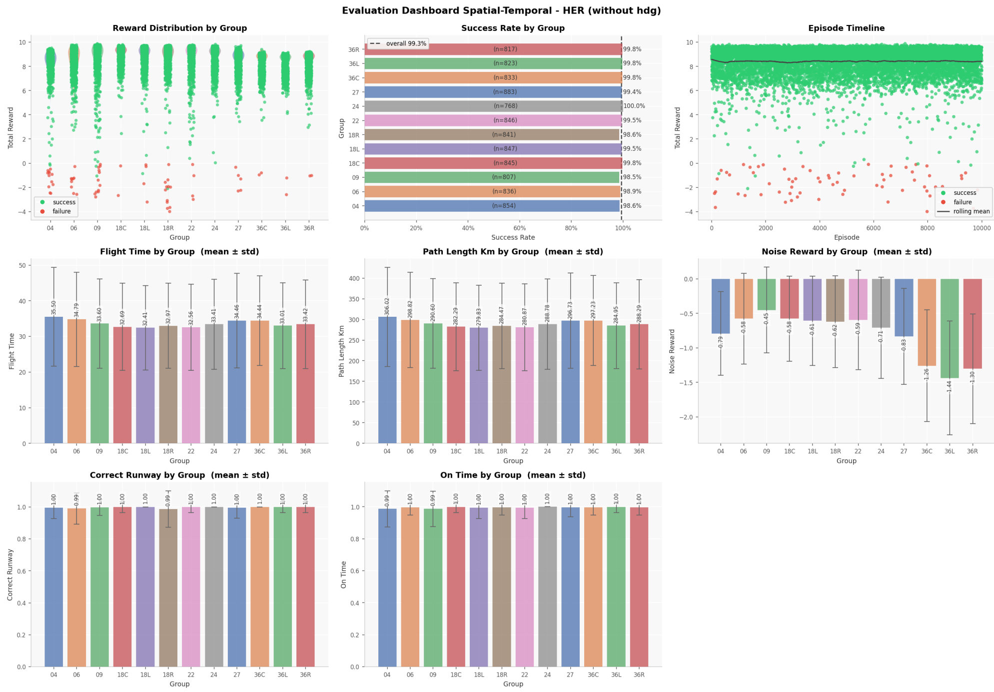


Fig. 15 Per-runway evaluation dashboard, Phase II spatial-temporal, HER without heading. Panels same as in Figure 14.

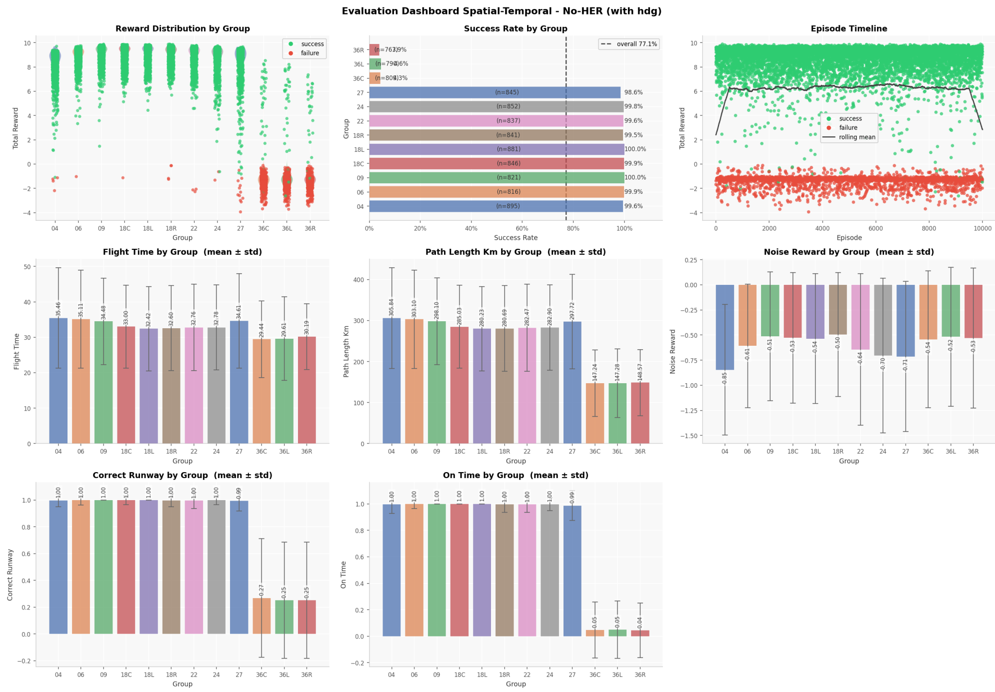


Fig. 16 Per-runway evaluation dashboard, Phase II spatial-temporal, No-HER with heading. Panels same as in Figure 14.

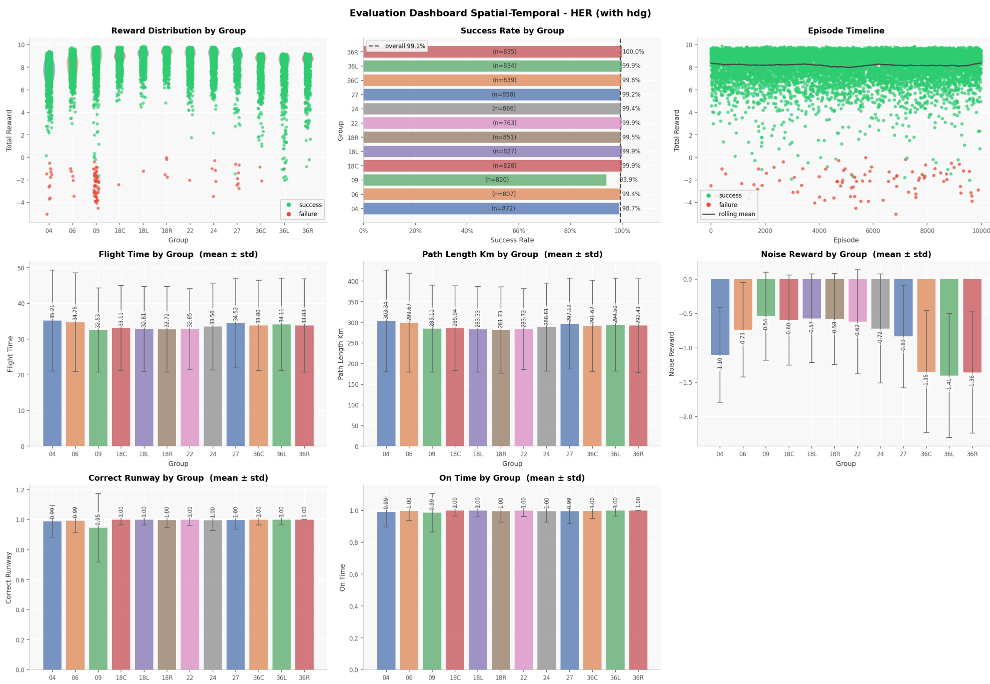


Fig. 17 Per-runway evaluation dashboard, Phase II spatial-temporal, HER with heading. Panels same as in Figure 14.

Acknowledgments

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